

Text Book of Food, Nutrition and Health

This book presents fundamental concepts of food, nutrition, and health in a way that is manageable for both instructors and students. With this in mind, it is my pleasure to write a brief foreword to the wonderful preparatory book **Text Book of Food, Nutrition and Health**.

I appreciate the authors for choosing to focus on foundational knowledge in clinical nutrition, particularly in the context of healthy eating.



Krishnasamy Shenkani is an expert in biological research across various fields of zoological studies. The author is currently serving as an Assistant Professor of Zoology at J. K. K. Nataraja College of Arts and Science, Komarapalayam, Namakkal District, and has nine years of experience as a postgraduate biology teacher at various colleges.



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Press**

Shenkani Krishnasamy
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and Health**

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Cover image: www.ingimage.com

Publisher:

Scholars' Press

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Dodo Books Indian Ocean Ltd. and OmniScriptum S.R.L publishing group

120 High Road, East Finchley, London, N2 9ED, United Kingdom

Str. Armeneasca 28/1, office 1, Chisinau MD-2012, Republic of Moldova,
Europe

Managing Directors: Ieva Konstantinova, Victoria Ursu

info@omniscryptum.com

Printed at: see last page

ISBN: 978-620-0-69600-7

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TEXT BOOK OF FOOD AND NUTRITION AND HEALTH

(A Revised Syllabus based on Periyar University)

YEAR - 2025

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FOREWORD

The book adheres to the scope and sequence of most one major course nationwide. also strive to make clinical nutrition, as a discipline, interesting and accessible to students. in addition to a comprehensive coverage of core concepts and foundational research, incorporated features that draw learners into the discipline in meaningful ways. this book provides a fundamental concept of food, nutrition and health with a scope that is manageable for instructors and students alike with this; it is my pleasure to write a brief foreword to this wonderful preparatory book **“THE TEXT BOOK OF FOOD, NUTRITION AND HEALTH”**. I am appreciating the authors to chosen this book chapter on basic preparatory knowledge of clinical nutrition. on the aspect of healthy nutrition include number of books are available in the form of, question bank for competitive examinations etc. except from this for the student community limited numbers of this like this book are available.

UG or PG level lack of syllabus covered book are available, this book might be useful to basic and advanced learner of clinical nutrition. written of this book contain simple language and easy to understand a basic learner and offers a valuable guide to the intellectual. this book can help teachers develop a shared vision and understanding of basic concepts of clinical nutrition and food, nutrition and health and techniques of biology and its flexible uses across subjects and grade levels.

This book has been created with several goals in mind, accessibility, customization, and student engagement all while encouraging students toward high levels of academic scholarship. instructors and students alike will find that this book offers a strong introduction to clinical nutrition in an accessible format.

ACKNOWLEDGEMENT

Words find no bounds to express my sincere thanks to all who made this Book to come true. my sincere thanks to **Capt.Dr.M. NALINI, M.Sc., M.Phil., Ph.D.**, Principal, J.K.K. Nataraja College of Arts and Science (Autonomous) Komarapalayam-638183, Namakkal Dt for extending the facilities to undergo this Book work.

I thank **Dr.S. UMAVATHI**, M.Sc., Ph.D., Assistant professor and head, PG & Research department of zoology, J.K.K. Nataraja College of arts and science, Komarapalayam, Namakkal Dt.

My specific thanks to **DR.S.V. RAMESH**, M.com., MPhil., Ph.D., Assistant professor, PG & Research department of commerce, J.K.K. Nataraja college of arts and science, Komarapalayam, Namakkal Dt. for her kind co-operation and encouragement to complete this manual successfully.

I wish to express my profound thanks to all the staff members, PG & Research department of zoology, J.K.K. Nataraja college of Arts and science, Komarapalayam, Namakkal Dt. I also wish to thanks assistant professor of the department of zoology.

My special thanks my research guide to **Dr.C. KANDEEPAN**, head and Associate professor of zoology, A.P.A. college of arts and culture, Palani. For helping me in many ways for completing this book works with great pleasure.

Last but not least i wish to extend my sincere thanks to all my friends and relatives for their co-operation and support in this manual to make success.

Dr. K. SHENKANI

Dr. S. KALAIMANI

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UNIT-I- NUTRITION AND DIETARY NUTRIENTS

BASIC CONCEPTS OF FOOD

The term 'food' refers to anything that we eat and which nourishes the body. It includes solids, semi-solids and liquids. Thus, two important features for any item to be called food are:

- (i) It should be worth eating, that is, it should be 'edible'.
- (ii) It must nourish the body.

FUNCTIONS OF FOOD:

There are basically three important functions of food:

1. Social Function: Food and eating have significant social meaning. Sharing food with any other person implies social acceptance. Food is also an integral part of festivity everywhere in the world. Have you noticed that certain occasions such as birth of a child or a marriage or birthdays, are celebrated by having feasts and serving delicacies? Food also has a specific significance and meaning in the religious context.

2. Psychological Function: We all have emotional needs, such as need for security, love and affection. Food is one way through which these needs are satisfied. For example, how do you feel when your mother prepares your Favorite food or dish? You feel that she loves you and cares for you. Food is often served as a reward also. Similarly, certain foods become associated with sickness, such as khichri and bland foods. Sickness is an unpleasant experience, hence, even the food items served during this state may be associated with unpleasant feelings.

3. Physiological Function: There are three physiological functions performed by food. These are energy giving, body building, regulating body processes and providing protection against diseases.

- (i) Food provides energy
- (ii) Food helps in body building
- (iii) Food regulates body processes and provides protection against diseases

1. COMPONENTS OF FOOD

Stable growth and nourishment depend on food. Food and nutrition are related. Food is the substance that nourishes the human body. Different foods have various nutrients that one can consume. Eating well-balanced meals can improve health and productivity. However, despite eating nutritious foods, it is frequently observed that the body does not respond as intended. In other words, meals that do not properly nourish the human body and are not effectively absorbed in the human body are not regarded as nutritious foods. Animals and plants can both serve as food sources.

Foods contain both energy- and non-energy-producing ingredients. The cells require calories, which are the source of energy. The following are some of the components that provide energy:

Carbohydrates

Carbohydrate is the human body's main source of energy. It is made up of sugars included in the food that immediately affect blood sugar levels.

Sources

Both plant and animal sources contain carbohydrates. Carbohydrates can be found in a variety of foods, including rice, wheat, cereals, millet, pulses, nuts, roots, potatoes, flour, molasses, sugar, sago, grapes, corn, bananas, mangoes, watermelons, vegetables, carrots, oranges, dates, apples, barley, goat liver, cow's and buffalo's milk, etc.

Classification

The composition indicates that there are three different forms of carbs.

- **Monosaccharide:** Monosaccharides are carbohydrates that only contain one molecule. These forms of carbohydrates include galactose, fructose, and glucose. Fruits, sweets, and honey all contain glucose. Fruits and root vegetables contain fructose. In milk, galactose can be discovered.
- **Disaccharide:** Disaccharides are types of carbohydrates that contain two molecules. like maltose, lactose, and sucrose. Milk includes lactose, while sugarcane, molasses, sugar, and date juice contain sucrose. Germinated seeds and sugar also contain maltose.
- **Polysaccharides:** Polysaccharides are a class of carbohydrates made up of many molecules. These carbohydrates include starch, glycogen, and cellulose, for instance. Starch is abundant in foods like rice, wheat, potatoes, sweet potatoes, etc. Fruits, vegetables, fruit peels, and seeds all contain cellulose.

Function

Carbohydrates have a significant part in human nutrition.

- Carbohydrates are primarily used by the body to generate heat and energy.
- Fats are burned with the aid of carbohydrates.
- Foods made with cellulose cure constipation in people.
- The body stores extra carbohydrates as the compound glycogen.
- These chemical compounds offer protection from ketosis.

Proteins

Protein is an essential component of the body that promotes the repair, growth, and reproduction of our damaged tissues and organs.

Sources

Fish, meat, eggs, milk, pulses, soybeans, beans, wheat, and other foods include proteins. There is hardly any protein in cereals, millet, potatoes, green vegetables, or fruits.

Function

- Proteins help the body perform its initial and most important job.
- The requirement for necessary amino acids is one of the protein's roles.
- To replenish the body's depleting cells, proteins produce new ones.
- Proteins make up the majority of the body's bones, muscles, organs, systems, blood cells, etc.
- It is crucial for the body's synthesis of antibodies.
- When the body is low on fats and carbohydrates, it breaks down stored protein to provide heat and energy.
- The blood's hemoglobin is composed of proteins.

Fats

With the help of critical fatty acids, fats are a crucial part of the freight system that carries many vitamins throughout the body.

Sources

Meat fat, butter, ghee, whole milk, ice cream, almonds, coconut oil, palm oil, cream, mustard oil, oilseeds, and other foods all contain fats.

Function

- Heat and energy are concentrated in fat. One gram of fat contains 9 kcal of energy.
- This organic substance offers vital fatty acids, which shield the body from skin ailments.
- The animal's body stores fat as a source of sustenance for the future.
- Vitamins A, D, E, and K are more easily dissolved and absorbed.
- It facilitates excreta evacuation by lubricating the rectum and anus.

Vitamins

Vitamins are those unique, organic nutrients that support healthy immunity and good body development.

Sources

Vitamin-rich foods include fish, including cod and halibut liver oil, whole milk, carrots, butter, ghee, vegetable oil, rice, flour, yellow pumpkin, mango, tomato, papaya, cabbage, and green vegetables and leaves.

Function

- Vitamins help the body grow normally, help make blood cells, and maintain healthy mucous membranes and nerve cells, among other things.
- Rod cells are created, immunity is boosted, and vitamins maintain gum health.
- It aids blood clotting and maintains an average level of prothrombin.
- Vitamins support fetal development, the production of breast milk, and other processes.
- It maintains healthy, silky skin.
- Vitamins support healthy retinal structure and vision.

Minerals

Minerals are a crucial part of the food. The body needs a variety of minerals to stay healthy. The body contains roughly 24 different types of minerals.

Sources

Mineral-rich foods include dairy, fish, meat, green vegetables, eggs, legumes, nuts, grains, liver, cod liver oil, seaweed, table salt, cheese, and fish liver oil.

Function

- Minerals support nutritional upkeep and bone development.
- These components aid in the development of teeth.
- Minerals aid in the synthesis of energy from fats and carbs.
- Red blood cells are produced with aid from minerals.
- Minerals keep the equilibrium of water within and outside of cells.
- It promotes muscle growth.
- Minerals impact the human body's physical and mental development and the average growth of bones, skin, and hair.

Water

Water is a necessary and significant part of the diet. Without water, the body cannot form, and its internal processes cannot work. Water makes up roughly 60–70% of the body weight. Every organ in the human body, including the blood, flesh, nerves, teeth, and bones, is formed with the help of water.

Function

- Water is necessary for the body's digestion of meals.
- All of the chemical reactions in the body require water.
- Water aids in transporting the body's essential components to each cell.
- Constipation is eliminated by water.
- Body temperature can be managed with the aid of water.

NUTRIENTS OF FOOD

Nutrition science is defined by Robinson as: "The science of foods, nutrients and other substances therein; their action, interaction and balance in relation to health and disease; the process by which an organism ingests, digests, absorbs, transports and utilizes nutrients and disposes off their end products". Nutrition is the process by which food is taken in and utilized by the body.

NUTRIENTS AND THEIR FUNCTIONS

Nutrients are the chemical substances present in food and are responsible for nourishing the body.

Nutrients are of two types:

1. Macronutrients
2. Micronutrients

Both macronutrients and the micronutrients are equally essential for good health. Each nutrient plays a significant role in the body.

Macronutrients: These are present in large quantities in foods and are also required in large amounts by the body: Carbohydrates, proteins, fats and oils are macronutrients

A. Carbohydrates

(i) Available carbohydrates: Carbohydrates are present in a large quantity as starch in cereals, legumes, pulses and potatoes. They are present as simple carbohydrates in sugar, jaggery, fruits, honey and milk. Starch and sugars are easily digested and provide energy to the body.

(ii) Unavailable carbohydrates or dietary fibre: They are present in the form of cellulose and hemicellulose which are not digested in our body. They add bulk to the stool and help in easy defecation process. Energy can be derived from carbohydrates, fats and proteins and it is measured in kilo calories. However, carbohydrates are cheapest sources of energy. If there is a short supply of carbohydrates and fats in our body, proteins are utilized for energy production. Function of proteins is to provide for body building. Therefore, carbohydrates have to be consumed in proper amounts to spare proteins for body building purpose.

Functions of carbohydrates: -

- Carbohydrates provide energy
- Carbohydrates are the main source of energy
- Carbohydrates spare proteins for body building function

Dietary fiber increases the bulk in stool and helps in defecation 1 gm of carbohydrate gives 4kcal of energy. Kilocalorie is the measure of energy in food.

Food sources of carbohydrates are:

Cereals - wheat, rice, bajra, maize, etc. Pulses - Rajma, channa, all dals'
Roots and tubers - potatoes, sweet potatoes, beetroot and tapioca Sugar, jaggery

B. Proteins

- Protein is needed in the body for body building.
- 1 gm of protein gives 4kcal of energy. Proteins are made up of smaller units known as amino acids. There are all together 22 amino acids, out of which there are 8 amino acids which our body cannot manufacture.
- Rest of the amino acids can be manufactured by the body. Essential amino acids are those which our body cannot manufacture and hence have to be supplied through the diet.
- Non-essential amino acids are those amino acids which our body can manufacture. Functions and sources of proteins

Functions

- (i) Needed for growth, maintenance and repair of tissues.
- (ii) Necessary for production of enzymes, hormones, antibodies, hemoglobin, etc.
- (iii) Help in the clotting of blood
- (iv) Provide energy, if necessary, Sources - Meat, poultry, fish, eggs - Milk, cheese, paneer, curd - soybeans, peas, pulses, - cereals, nuts and oilseeds like groundnuts, etc.

Special features

- (i) Animal proteins, i.e., proteins from meat, eggs, milk, etc., are better than vegetable proteins, i.e., proteins, from pulses, cereals, etc. This is because proteins from vegetable sources do not contain all essential amino acids.
- (ii) Including two or more sources of vegetable proteins in each meal helps to improve the quality of proteins and their utilization.

C. Fats and Oils

Fats and oils are the concentrated source of energy in our diet. 1 gm of fat gives 9 kcal of energy. Fats are made up of small units called fatty acids. The nature of fats is dependent on the type of fatty acids present. Fatty acids may be saturated or unsaturated. Saturated fatty acids are found in solid fats whereas oils contain more of unsaturated fatty acids.

Vegetable oils are rich in unsaturated fatty acids. Do you know there is a difference between fats and oils? If a substance is liquid at room temperature it is called oil and if it is solid at the room temperature, it is known as fat.

FUNCTIONS AND SOURCES OF FATS AND OILS:

- (i) Provide concentrated source of energy
- (ii) Reduce the use of proteins for energy
- (iii) Carry fat soluble vitamins (A, D, E, K) into the body and help in the absorption of these vitamins

- (iv) Help to maintain body temperature. The layer of fat under the skin helps to conserve body heat
- (v) Act as a cushion to certain vital organs
- (vi) Help in growth of tissues Sources - Cooking oils, ghee, butter - Oilseeds, nuts - Meat, poultry, fish, eggs - Whole milk, cheese

Special features

- (i) Fats improve the texture as well as absorb and retain flavors making meals more appetizing.
- (ii) Fats have properties that help them to remain in the stomach longer and prolong the feeling of fullness.

1. Micronutrients

Other important nutrients which are present in small quantities in foods but are essential for our body are called micronutrients. These are minerals and vitamins and are required in very small quantities. If these micronutrients are not eaten in required amounts, it results in deficiency diseases.

Minerals and vitamins are called micronutrients.

1. Vitamins

Our body contains very little quantity of vitamins; however, you will be surprised to know that they are responsible for all the major functions of the body. These vitamins are of two types:

- (i) Fat soluble: A, D, E and K (ii) Water soluble: B and C

2. Minerals

Minerals constitute a very small amount of the total body tissues. However, these are essential for many vital processes and also for the maintenance of the body. In total, there are about 19 minerals required by the body in various amounts. Let us now study some of the important minerals.

Calcium:

Calcium and phosphorus are available in sufficient quantities in milk, curd, green leafy vegetables, ragi and oil seeds. Other foods also provide fair quantity of calcium. The major function of calcium is the formation and development of bones and teeth. Calcium is also required in blood clotting and muscular contraction. Calcium is necessary for bone formation, blood clotting and muscular contraction. Deficiency of calcium in the body results in poor bone development, particularly in children, women and elderly. The deficiency disease is known as osteoporosis. In this, the bones become brittle and people become prone to frequent fractures.

Iron:

Iron is required in very small quantity by the body. It is an important material present in hemoglobin which is a part of red blood cells and is responsible for the red colour of blood. Whole grain cereals and pulses are the major sources of iron in our diet. Other sources of iron are green leafy vegetables, egg yolk, liver and meat. In our country, majority of the population, especially women and children, suffer from iron deficiency disease called anemia. Young girls (12-18 yrs.) need more iron rich foods in

their diets because of loss of iron during the menstrual cycle. Extra iron is also needed during pregnancy for healthy development and growth of the fetus. This is not because people do not consume food which are rich in iron but because the absorption and utilization of iron is poor. This is due to the presence of certain naturally occurring constituents in food called oxalates and phytates. These oxalates and phytates are called inhibitors of iron. Vitamin C and proteins help in better absorption of iron and are known as enhancers of iron. It is essential for hemoglobin formation.

Iodine:

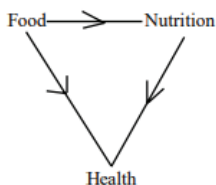
Iodine is an important substance present in thyroxine hormone produced from thyroid gland. Thyroxine regulates various functions of the body. We get iodine from water and food. The foods which grow in iodine rich soil provide iodine for us. Sea foods are also rich in iodine. Iodine deficiency disorder is known as goiter or enlargement of the neck region. Deficiency of iodine causes mental retardation in children. Recent studies have shown a direct link between iodine deficiency and academic performance of children. Iodine deficiency disorders have been identified in many parts of India.

To avoid goiter, we must have iodine rich food sources in our daily meals. Iodized salt is a good source of iodine and we must consume it instead of the non-iodized salt. Certain foodstuff like cabbage, cauliflower, radish, ladies' finger, oilseeds etc., contain substances known as goitrogens which interfere with the body's ability to produce and use thyroxine. These goitrogens are destroyed on cooking. Therefore, these foodstuffs should be cooked before eating.

INTER-RELATIONSHIP OF FOOD, NUTRITION AND HEALTH

Nutrition is a scientific discipline in which food is a major focus of interest. The simplest definition of nutrition is the study of what happens to food once it enters the mouth and thereafter. A more formal definition of nutrition is study of processes by which the living organism receives and utilizes the materials necessary for growth, renewal and maintenance of body components. All foods contain some essential substances which perform important functions in our body. These essential substances contributed by our food are called nutrients. These nutrients help us to maintain our body functions, that is, to grow and to protect our organs from diseases and infections.

The health of a person depends on the type and quantity of food stuff consumed. Good nutrition is essential for a person to grow and develop normally and to remain healthy throughout life. When a person does not eat proper food, there are chances of the body not developing normally. There are chances that some organs of the body may start malfunctioning or there may be some disease. Poor nutrition may also influence the mental and social wellbeing adversely.



CONCEPT OF BALANCED DIET

A balance diet can be defined as the one which contain different types of food in such quantities and proportion that calories, energy and other requirements are adequately met and a small provision is made for extra nutrients to withstand short duration of illness. “A balanced diet provides all the nutrients in required amounts and proper proportions”.

Balanced diet can be achieved by carefully selecting and combining food items from different food groups. The amount of food needed to meet the daily dietary requirements will differ according to age, gender, physiological status and level of physical activity. A balanced diet should have a variety of foods in moderation and proper proportions. Hence, balanced diet should meet these aspects:

1. A BALANCE DIET CONSISTS OF DIFFERENT ITEMS.

It includes a variety of food which ensures that all nutrients are supplied. This can be achieved by selecting items from each food groups (each food group supplies certain nutrients)

2. SELECTING ITEMS FROM DIFFERENT FOOD GROUP

Including items from each food group ensures that all the nutrients will be supplied.

3. A BALANCE DIET MEET THE NUTRIENTS NEEDS

Because of the amount and proportion of the food selected. This is based on the recommended dietary intakes laid down for the individual.

4. BALANCED DIET PROVIDE FOR PERIOD OF LEANESS (SAFETY MARGIN)

It is a safety margin or a little extra for that time when you do not need your nutrient needs adequately.

Goals of a Balanced Diet:

1. A balanced diet should provide around 50-60% of total calories from carbohydrate, 10-15% from protein and 20-30% of total calories from fat.
2. Calorie allowance can be +50, while for all other nutrients minimum RDA must be met.
3. Energy from cereals should not be more than 75 percent. Include two cereals in one meal; for e.g. rice and wheat.
4. Two to three servings of pulses should be taken a day.

5. To improve protein quality, the ratio of cereal protein to pulse protein should be 4:1. Include at least one medium size fruit. The fruit can be given raw without much cooking.
6. Five servings of fruits and vegetables should be included in a day.
7. Foods rich in fibre should be included.
8. The diet should include minimum 100ml milk per day.
9. One third of the nutritional requirements, at least calorie and protein should be met by lunch or dinner.

Components of a Balanced Diet

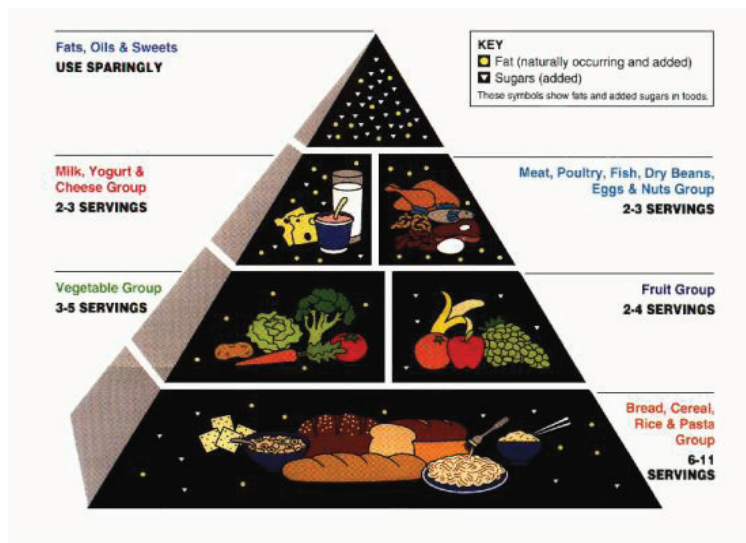
A balanced diet must provide the building blocks for growth and repair, and sufficient energy.

The components of a balanced diet are:

- 1) Carbohydrates- the majority of energy intake should come from these molecules, which provide an energy source.
- 2) Proteins- needed for growth and repair, and are minor energy source.
- 3) Fats- major energy source, used for insulation and cell membranes, hormones and cholesterol.
- 4) Vitamins- these play an important role in the chemical processes taking place inside cells; some vitamins are water-soluble and some are fat-soluble.
- 5) Minerals- these are the inorganic elements occurring in the body that are essential for its normal functions.
- 6) Water- an important transport medium. Water is essential for the body to function, and it is involved directly in some metabolic reactions.

FOOD PYRAMID

It is a triangle diagram, representing the optimal number serving to be eaten each day from each food group. Its purpose is to make healthy eating easier that is the correct amount of nutrients to maintain good health, or to get a balanced diet. The first pyramid was published in Sweden in 1970. In 1992 USDA (united state department of agriculture) introduced food guide pyramid as a general plan of what to eat each day. The food guide pyramid is a valuable tool for planning a health promoting diet. By incorporating the principle of balance, variety and moderation, an individual can still eat their favorite foods while following the food guide pyramid. A balanced diet should provide 50-60% of the calories from complex carbohydrates which is the major part of the diet thus making it the base of the food pyramid. Therefore 6-11 servings of cereals are recommended. Protective foods are essential in providing vitamins, minerals and fiber. Therefore 3-5 servings of vegetables and 2-4 servings of fruits form the next level of food pyramid. 2-3 servings of pulses, milk and milk products, egg, meat and fish form the next level of food pyramid to meet 10-15% of protein of the total calories. The tip of the food pyramid has sugars, fats and oils which are to be used sparingly since they are energy dense foods.



Source: Food guide pyramid, USD

It was updated in 2005 to “my pyramid” my pyramid helped them to choose food and amount that are right for them to balance with their physical activities. It is meant for any healthy person who is above the age of 2 years. Be active, the person climbing stairs and remind the person to be physically active. Vary your choices, the six bands stand for the five food groups plus an area for oil. For health it is essential to consume a variety along and within these groups to get the way of nutrients needed by the body. Number of food and food groups supplies all the nutrients, fiber and other substances the body needs. Besides, variety adds flavors, interest and pleasure to eat. Think in proportion the food groups bands differ in width, reminding consumers to eat more of some types of food than others. These widths are just estimates and not specifically the amount that is right for the person. Take in moderation: moderation is the key, mind-set and every day eating habits should adapt moderation for each food group.

The wider base stands for foods with little, no solid fats or added sugars, they should be eaten more often. The narrower top stands for foods with more of these foods he can consume. Customize: My pyramid is available for kids and also for vegetarians. They can be used to a person's need. Improve gradually: small steps should be taken towards healthier eating and active living. Small steps add up to big benefits. No food contains all the essential nutrients, eating food from each of the food groups ensures that all nutrient needs are met. For example, people who have low energy allowance are advised to eat energy food from all food groups. People with high energy allowances are advised to eat less-dense energy food. There is no substitute to balance diet. The principle of balance diet is needed to follow throughout the life.

NUTRITIONAL NEEDS OF INFANTS

Infancy is a period of rapid growth. The development during infancy is more rapid than during at any other period in the life time of an individual.

Growth and Development

The growth and development is accompanied by a number of physiological changes which include changes in body size and body composition, changes in the gastro intestinal system, excretory system and circulatory system. Nutrition is crucial and proper dietary modifications are vital during this period.

Immunization

Malnutrition during infancy leads to a higher incidence of infant mortality. Besides malnutrition, infection causes mortality. Immunization protects the children against disease. Table 6 below presents the immunization schedule for infants and children.

Good nutrition is essential for the growth and development that occurs during an infant's first year of life. As an infant's mouth, tongue, and digestive tract mature, the infant shifts from being able to only suckle, swallow, and take in liquid foods, such as breast milk or infant formula, to being able to chew and receive a wide variety of complementary foods.

Table 6 National Immunization Schedule (NIS) for Infants, Children

Age	Vaccine
Soon after birth	Hepatitis B 1 st dose, OPV 1 st dose, BCG
6 weeks	Hepatitis B 2 nd dose, DPT 1 st dose, OPV 2 nd dose
10 weeks	DPT 2 nd dose, OPV 2 nd dose
14 weeks	DPT 3 rd dose, OPV 4 th dose
6 weeks, 10 weeks and 14 weeks	OPV 5 th dose, Hepatitis B 3 rd dose
9 months to 12 months	Measles
15 to 18 months	MMR
16-24 months	DPT, OPV 1 st Booster
2 years	Typhoid vaccine
5 – 6 years	DPT, OPV 2 nd Booster

DPT- Diphtheria, Pertussis, And Tetanus; OPV-Oral Polio Vaccine; MMR- Measles, mumps, and rubella

Source: <https://mohfw.gov.in/sites/default/files/245453521061489663873.pdf>

Energy:

Infants need energy from food for activity, growth, and normal development. Energy comes from foods containing carbohydrate, protein, or fat. A general indicator of infant consuming an adequate kilocalories per day is the infant's growth rate in length, weight, and head circumference.

Carbohydrate

The major type of carbohydrate normally consumed by young infants is lactose, the carbohydrate source in breast milk. Lactose-free infant formulas, such as soy-based infant formulas provide carbohydrates in the form of sucrose. In later infancy, infants derive carbohydrates from additional sources including cereal and other grain products, fruits, and vegetables.

Protein

Breast milk and infant formula contains protein. The complementary foods such as meat, poultry, fish, egg yolks, cheese, yogurt, pulses, cereals and other grain products provide adequate protein.

Fat

Breast milk and infant formula are important sources of lipids, including essential fatty acids, during infancy.

Vitamin A

Breast milk and infant formula are major food sources of vitamin A. Additional sources of vitamin A or carotenes for infants consuming complementary foods include egg yolks, yellow and dark green leafy vegetables and fruits e.g., spinach, greens, sweet potatoes and liver.

Vitamin E

Infants receive vitamin E from breast milk and infant formula. Other vitamin E sources for older infants include green leafy vegetables, vegetable oils and their products, wheat germ, whole-grain breads, cereals and other fortified or enriched grain products, butter, liver, and egg yolks.

Vitamin K

Sources of vitamin K include infant formula, green leafy vegetables, pork, and liver.

Vitamin C

Breast milk and infant formulas are major food sources of vitamin C. Additional vitamin C sources include vegetables (e.g., tomatoes), fruits (e.g., citrus fruits, papaya, and strawberries), and regular fruit and vegetable juices which are naturally high or fortified with vitamin C.

Vitamin B₁₂

An infant's vitamin B₁₂ stores at birth generally supply his or her needs for approximately 8 months. Infants consuming appropriate amounts of breast milk from mothers with adequate B₁₂ stores or infant formula receive adequate amounts of this vitamin. Complementary foods such as meat, egg yolks, and dairy products provide this vitamin later in infancy as well.

Calcium

An infant can obtain sufficient calcium by consuming adequate amounts of breast milk or infant formula. Older infants can obtain additional calcium from complementary foods such as yogurt, cheese, cottage cheese (paneer), fortified or enriched grain products, some green leafy vegetables (such as turnip and greens), and tofu.

Iron

Sources of iron for infants include breast milk, infant formula, meat, liver legumes, whole-grain breads, cereals, or fortified or enriched grain products, and

dark green vegetables. Heme iron is found primarily in animal tissues, including red meat, liver, poultry and fish and non-heme iron is found in breast milk, infant formula, cereals, or other grain products legumes, fruits and vegetables. Infants receive most of the iron in their diets as non heme iron.

Tofu, or bean curd, is a popular food derived from soya. It is a staple ingredient in Chinese cookery and is a good source of protein, containing all eight essential amino acids. It is also an excellent source of iron and calcium and the minerals manganese, selenium and phosphorous.

Zinc

Infants obtain zinc from breast milk, infant formula, meat, poultry, liver egg yolks, cheese, yogurt, legumes, and whole-grain breads, cereals, and other fortified or enriched grain products.

Sodium

Healthy, full-term infants consuming primarily breast milk or infant formula of standard dilution receive a relatively small amount of sodium but an amount adequate for growth.

Fiber

Breast milk contains no dietary fiber, and infants generally consume no fiber in the first 6 months of life. As complementary foods are introduced to the diet, fiber intake increases. Dietary fiber is found in legumes, whole grain foods, fruits, and vegetables.

Water

Infants' water needs are met from consuming breast milk, infant formula, and complementary foods. Water is also formed in the body in chemical reactions occurring to metabolize protein, fats, and carbohydrates. Under normal circumstances, the water requirements of healthy infants who are fed adequate amounts of breast milk or properly reconstituted infant formula are met by the breast milk or infant formula alone.

DIETARY PATTERNS OF INFANTS

Dietary patterns of infants refer to the types and timing of foods introduced to babies during their early growth and development. These patterns significantly influence their nutritional status, growth, and long-term health outcomes. Below is an overview:

Stages of Infant Dietary Patterns

1. Exclusive Breastfeeding (0-6 Months)

- Breast milk or formula is the sole source of nutrition.
- Provides all necessary nutrients, immune support, and hydration.
- WHO recommends exclusive breastfeeding for the first six months.

2. Complementary Feeding (6-24 Months)

- Gradual introduction of solid foods alongside breastfeeding/formula.
- Foods include mashed fruits, vegetables, cereals, and protein sources like lentils or pureed meat.

Key considerations:

- **Nutrient-dense foods:** Rich in iron, zinc, calcium, and vitamins.
- **Texture:** Start with purees and progress to soft, small chunks.
- **Avoidance of allergens:** Introduce allergenic foods early under guidance.

3. Transition to Family Diet (12+ Months)

- Infants begin consuming small portions of family meals.
- Encourage balanced meals with fruits, vegetables, whole grains, and proteins.
- Limit salt, sugar, and processed foods.

Key Nutritional Considerations

- **Iron and Zinc:** Critical after 6 months as stores from birth start depleting.
- **Healthy Fats:** Support brain development (e.g., avocado, nuts in paste form).
- **Vitamin D:** Supplementation may be necessary if breastfeeding or low sun exposure.
- **Hydration:** Gradual introduction of water after six months.

Common Challenges

- **Food allergies:** Monitor for reactions like rashes, vomiting, or diarrhea.
- **Picky eating:** Maintain patience; reintroduce foods multiple times.
- **Cultural and economic factors:** Influence food choices and patterns.

Good Practices for Healthy Dietary Patterns

1. Follow responsive feeding—recognize hunger and satiety cues.
2. Avoid force-feeding or using food as a reward/punishment.
3. Create a positive mealtime environment.
4. Encourage self-feeding to build motor skills and autonomy.
5. Regularly consult pediatricians for guidance on growth and nutrition.

NUTRITIONAL NEEDS OF PRESCHOOL AND SCHOOL GOING CHILDREN

The rapid growth during infancy is followed by a generally slow growth between one to six years. The child becomes more active and the social and environmental influences have a great impact on their food behavior and eating pattern. The need for nutrients is increased as growth and development continue.

Growth Pattern

During the second year, the increase in height is about 10 cm and weight gain is 2 to 2.5 kg. After two years annual gain in height and weight is only 6 to 7 cm and 1.5 to 2 kg respectively. However, there is a wide variance in the physical development of children.

As growth proceeds, changes occur in a) proportion of water, b) muscle tis-sue, c) fat deposits and d) skeletal structure. The body water gradually decreases and there is addition of adipose tissue and minerals to the bones.

Nutritional Needs During Preschool

Maintaining a balanced diet and regular exercise is important for all individuals, especially school-aged children (6-12 years). These children are required to eat a variety of foods from each food group to ensure optimal intake of all vitamins and minerals. At the same time, they may face new challenges regarding food choices and habits. Decisions about what to eat are partly determined by what is provided in school, at home, the influences from friends at school, and the media, especially television.

Poor nutrition compromises both the quality of life of school-aged children but also their potential to benefit from education. Attaining optimal nutrition involves eating three meals a day and two nutritious snacks, as well as limiting the intake of high sugar and high fat foods. Consuming generous amounts of fruits, vegetables, lean meats and low-fat dairy products, including three servings of milk, cheese or yoghurt to meet their calcium requirement, can also prevent many medical problems. This includes becoming overweight, developing weak bones, and developing diabetes.

Adequate nutrition of school aged children will also ensure they grow to their full potential, and provide the stepping stones to a healthy life.

Essential nutrients for the school-aged child

School-aged children grow significantly, but at slower rate, whilst being very physically active in general. As a result, their nutritional needs are high and critical. Additionally, genetic background, gender, body size and shape are all important determinants of nutrient requirements. A recent review of research on the effects of deficiencies in zinc, iodine, iron and folate on the cognitive development of school-aged children showed that nutrition has an impact on children's ability to think. For example, deficiencies in iron and zinc have been

associated with impairment of neuropsychological function, retardation of growth and development, reduced immunity and increased vulnerability to infectious diseases. The essential nutrients for optimal health are:

Energy

Carbohydrates and fats provide energy for growth and physical activity. During periods of rapid growth, appetites increase and children tend to eat constantly. When growth slows, appetites diminish and children eat less at meal times. The brain needs energy to function properly and hence the supply of glucose is relevant and critical. Cognitively demanding tasks, such as schoolwork, require regular supplies of glucose to the brain in order to enhance cognitive functioning and improves memory and mood.

Protein

Protein builds, maintains and repairs body tissue. It is especially important for growth. It's important that parents encourage children to eat two to three servings of protein daily. Good sources of protein for children include meat, fish, poultry, milk and other dairy.

Essential fatty acids

Deficiency of unsaturated fatty acids may have a negative impact school performance. In a randomised controlled trial, six months of treatment with fatty acid supplements among 102 dyslexic school aged children significantly improved reading age on standardised tests of single word reading. Dyslexia and Dyspraxia are associated with difficulties with spelling, handwriting and written expression.

Calcium

Calcium is important in building strong bones and teeth. Bone density suffers when calcium needs are not met during childhood years. Osteoporosis, a weakened bone disease affects a significant proportion of adults. This begins in childhood if diets are not providing adequate calcium-rich foods. Milk and dairy products and from some dark green, leafy vegetables are good sources of calcium.

Iron

Children need iron because of rapidly expanding blood volume during growth. Meats, fish, poultry, and enriched breads and cereals are the best sources of dietary iron.

Nutritional advice for school-aged children

The best nutrition advice to keep your children healthy includes encouraging them to eat breakfast every day to help maintain concentration in class. A good breakfast should be able to provide a third of the total daily energy requirement. A typical breakfast includes a cereal (for example, rice, bread, and

oats), protein-rich food such as egg, a glass of milk and vitamin C rich fruit e.g. orange, and papaya. One study found that an overnight and morning fast among school children had deleterious effects on memory and attention. Empirical evidence from research on the effects of breakfast on cognition shows that particularly for younger children, skipping breakfast can have adverse effects on both general energy levels and cognition of school children.

- Eat a variety of foods in order to have adequate nutrient intake.
- Balance the food you eat with physical activity.
- Choose a diet with plenty of grain products, vegetables and fruits.
- Choose a diet low in fat, saturated fat, and cholesterol. Buy low-calorie and low-fat meals, snacks and deserts, low fat or skim milk and diet drinks.
- Choose a diet that provides enough calcium and iron to meet their growing body's requirements.
- Teach children from an early age about nutrition, foods, drinks, healthy eating and drinking. They should be taught what happens to the food they consume.
- Choose a diet moderate in sugars and salt. Avoid giving large amounts of sweet deserts, soft drinks, fruit-flavored drinks, sugar-coated cereals, chips or candy, as they have little nutritional value.

DIETARY PATTERNS FOR SCHOOL AGED CHILDREN

The dietary patterns of school-aged children (6-12 years) play a pivotal role in their growth, cognitive development, and overall health. This period is critical for establishing lifelong healthy eating habits.

Balanced Nutrition Needs

- Energy for growth and active lifestyle.
- Macronutrients: Carbohydrates, proteins, and healthy fats.
- Micronutrients: Essential vitamins and minerals.

Regular Meal Patterns

- Breakfast: Crucial for concentration and energy.
- Lunch and Dinner: Balanced meals with all food groups.
- Snacks: Healthy options to meet energy demands.

Hydration

- Adequate water intake; limit sugary beverages.

Cultural and Peer Influences

- Food choices influenced by culture, family habits, and peers.

HEALTHY EATING PRACTICES

1. **Balanced Meals**

- Include all food groups in meals.
- Emphasize fruits, vegetables, whole grains, and proteins.

2. **Healthy Snacking**

- Examples: Fruits, nuts, yogurt, boiled eggs.
- Avoid chips, candies, and sugary drinks.

3. **Limit Processed Foods**

- Reduce intake of high-sugar, high-salt, and trans-fat-rich foods.

NUTRITIONAL REQUIREMENTS OF PREGNANT AND LACTATING WOMEN

Nutrition requirements increase tremendously during pregnancy and lactation owing to the physiological changes. Wholesome nourishment before pregnancy has a greater impact on the long-term health on the mother and foetus. A well-nourished foetus enters life with good physical and mental health.

Physiological Changes in Pregnancy

Foetal development is accompanied by many physiological, biochemical and hormonal changes which influence the nutrient needs and the efficiency with which the body utilizes them. The changes include

i. Increased basal metabolic rate (BMR)

Due to foetal growth the BMR increases.

ii. Gastro intestinal changes

Gastro intestinal motility diminishes which may result in constipation.

iii. Hormonal changes

During pregnancy, there is increased secretion of the following hormones: Aldosterone, Progesterone, Thyroxine and Parathormone.

iv. Changes in body fluid

The blood volume expands by 50 per-cent and this increased amount of blood is required to carry nutrients to the foetus and remove metabolic wastes from the foetus.

v. Altered renal function

Increased blood volume and increased production of waste products like creatinine, urea and other wastes due to foetal and maternal metabolism produces a high glomerular filtration rate (GFR).

vi. Weight Gain during Pregnancy

Less than half of total weight gain resides in the foetus, placenta and amniotic fluid. Women with desirable body weight is 12.5 kg ranging between 11-13 kg.

NUTRITIONAL REQUIREMENTS

In addition to the RDA for an adult woman the nutritional needs increase during pregnancy.

Energy

Energy requirement during pregnancy is increased because of the additional energy required for growth and activity of foetus, growth of placenta and maternal tissues, increase in maternal body size and steady rise in BMR.

Protein

An additional protein intake is essential for:

- Growth of the foetus.
- Development of placenta
- Enlargement of uterus, mammary glands
- Increased maternal blood volume
- Formation of amniotic fluid

Fat

ICMR expert committee has suggested an intake of 30g of visible fat/day during pregnancy.

Calcium

Additional calcium is needed for the growth and development of bones as well as teeth of the foetus and also for the protection of calcium resources of the mother to meet the high demand of calcium during lactation.

Iron

The requirement of iron increases from 21mg/day to 35mg/day during pregnancy. The increased requirement is due to I. expansion of maternal tissues including red cell mass, iron content of placenta and blood loss during parturition. I. to build the iron store in foetal liver to last for at least 4-6 months after birth. This is because the baby's first food milk is deficient in iron.

Iodine

Due to increase in BMR, iodine needs are also enhanced during pregnancy.

Zinc

Deficiency of zinc adversely affects the outcome of pregnancy. Zinc deficiency leads to foetal mortality, foetal malformations and reduced intra uterine growth rate.

Sodium

Normal sodium intake without restriction is advised during pregnancy. Sodium is restricted when there is oedema or hypertension.

Vitamins

An additional allowance of Vitamin A is needed during pregnancy. Vitamin D is essential as it enhances maternal calcium absorption. Vitamin K is required for synthesis of prothrombin which is essential for normal coagulation of blood. A liberal vitamin K level in the mother's blood is helpful in preventing

neonatal hemorrhage. Vitamin C, pyridoxine, and vitamin B12 needs are increased during pregnancy.

Folic acid

Folic acid is essential for increased blood formation i.e. hematopoiesis and for synthesis of essential components of DNA/RNA which increase rapidly during growth.

Problems during Pregnancy

Nausea and vomiting, Constipation, Heart burn, Oedema and leg cramps, Pica, Anemia, Pregnancy induced hypertension (PIH) and Gestational diabetes.

DIETARY PATTERNS OF PREGNANT WOMEN

Pregnancy is a crucial phase where dietary patterns significantly influence maternal and fetal health. A balanced diet supports the baby's development and reduces pregnancy-related complications for the mother.

Key Nutritional Requirements

1. Macronutrients

- **Carbohydrates:** Energy source (whole grains, fruits, vegetables).
- **Proteins:** Vital for fetal growth and maternal tissue repair (eggs, chicken, fish, dairy, lentils).
- **Healthy Fats:** Promote brain and nervous system development (avocado, nuts, seeds, fatty fishlike salmon).

2. Micronutrients

- **Folic Acid:** Prevents neural tube defects (spinach, broccoli, fortified cereals, supplements).
- **Iron:** Supports increased blood volume and oxygen transport (meat, spinach, beans, fortified cereals).
- **Calcium & Vitamin D:** Strengthens fetal bones and teeth (milk, yogurt, fortified orange juice, sunlight for Vitamin D).
- **Iodine:** Ensures proper brain development (iodized salt, dairy, seafood).
- **Zinc:** Supports immune function and cell growth (nuts, seeds, whole grains).
- **Vitamin C:** Enhances iron absorption (citrus fruits, strawberries, bell peppers).

3. Hydration

- Aim for 8-10 glasses of water daily to maintain amniotic fluid levels and prevent dehydration.

Dietary Guidelines by Trimester

1. **First Trimester** (0-12 weeks)
 - Focus on folic acid, iron, and hydration.
 - Small, frequent meals to manage nausea.
2. **Second Trimester** (13-26 weeks)
 - Increase calorie intake moderately (~300 extra calories/day).
 - Prioritize protein, calcium, and iron-rich foods.
3. **Third Trimester** (27-40 weeks)
 - Emphasize energy-dense foods and healthy fats to support rapid fetal growth.
 - Manage meal sizes to reduce heartburn.

Healthy Eating Practices

1. **Balanced Diet**
 - Include a variety of fruits, vegetables, lean proteins, whole grains, and dairy.
 - Limit processed foods, added sugars, and trans fats.
2. **Meal Frequency**
 - Eat small, frequent meals to manage hunger, nausea, and energy levels.
3. **Safe Food Choices**
 - Avoid raw/undercooked meats, fish high in mercury (shark, swordfish), unpasteurized dairy, and raw eggs.
4. **Supplements**
 - Prenatal vitamins for folic acid, iron, calcium, and Vitamin D.

COMMON CHALLENGES AND SOLUTIONS

1. **Morning Sickness**
 - Eat bland, easily digestible foods (crackers, toast).
 - Sip ginger tea or lemon water.
2. **Heartburn**
 - Avoid spicy and acidic foods.
 - Eat smaller meals and avoid lying down after eating.
3. **Constipation**
 - Increase fiber intake (fruits, vegetables, whole grains) and hydration.
4. **Pica (Cravings for Non-Food Items)**
 - Seek medical advice to address underlying deficiencies.

Special Considerations

1. **Cultural and Economic Factors**
 - Adapt meal plans to include affordable, culturally relevant foods.
2. **Weight Management**
 - Gradual weight gain based on pre-pregnancy BMI (underweight, normal, or overweight).
3. **Physical Activity**
 - Light exercise, like walking or prenatal yoga, complements dietary patterns.

NUTRITIONAL REQUIREMENTS DURING LACTATION

Adequate nutrition for the mother during lactation is also of vital importance as the infant is dependent on mother's milk for its nutrition for the first few months of life. Inadequate nutrition during lactation is reflected on both the quality and quantity of milk secreted.

Physiology of lactation

The Table 13 below gives the summary of hormonal control of lactation.

NUTRIENT NEEDS DURING LACTATION

ICMR nutrient recommendations for a lactating mother are based on the composition of breast milk and the fact that 850 ml of milk is produced daily. However, the milk secretion continues to increase in the early periods of lactation up to six months and then gradually decreases. Therefore, the nutrient requirements are given for the two periods in lactation i.e. 0-6 months and 6-12 months.

Energy

The lactating mother requires additional energy for the production of milk. Based on the optimal output of 850 ml/day, the additional allowance is recommended during first six months of lactation.

Protein

Due to production of milk, protein requirement also increases.

Fat

The total fat in breast milk is not influenced by the mother's diet. The fat also provides energy density to meet the higher energy requirement during lactation.

Calcium

The requirement for calcium doubles during lactation.

Iron

Since most mothers have lactational amenorrhea, it results in saving of nearly 1mg iron per day which otherwise would have been lost due to menstruation. There is a marginal increase in the iron intake.

Table 13 Summary of Hormonal Control of Lactation

S. No.	Hormone	Source	Function
1	Estrogen	Ovary	Stimulates breast development
2	Progesterone	Placenta	Prepares breast for milk production by changing glandular cells to secreting cells
3	Prolactin	Anterior pituitary	Stimulates milk production
4	Oxytocin	Posterior pituitary	Facilitates release of milk from alveolus

Source: Srilakshmi (2011)

Vitamins

The additional need of vitamin A during lactation is based on the amount secreted in the mother's milk. As the calorie and protein requirements increase during lactation, the requirements of B vitamins also increase correspondingly. Ascorbic acid content increases during lactation.

Galactagogues

Galactagogues are foods that help to produce more milk. Garlic, milk and almonds are considered to increase milk production. Studies carried out on nursing mothers have revealed that extra amounts of body building foods like fish and mutton increase the secretion of breast milk. Lactating mothers are also given special preparations containing ajwain and fenugreek seeds, which supply iron, protein, calcium and B- complex vitamins.

DIETARY PATTERNS OF LACTATING WOMEN

- Lactation is a nutritionally demanding period as the mother provides essential nutrients to the baby through breast milk.
- A balanced diet ensures sufficient milk production, maternal health, and baby's growth.

KEY NUTRITIONAL REQUIREMENTS

1. Macronutrients

o Calories:

- Additional 450-500 kcal/day is needed for adequate milk production.
- Sources: Whole grains, fruits, vegetables, nuts, and seeds.

o Proteins:

- Vital for milk quality and maternal recovery (eggs, poultry, fish, legumes, dairy).

o Healthy Fats:

- Essential for baby's brain development (avocado, nuts, seeds, fatty fish).

2. Micronutrients

- **Calcium:** Supports maternal bone health and milk composition (milk, cheese, yogurt, leafy greens).
- **Iron:** Replenishes maternal iron stores lost during delivery (meat, spinach, lentils).
- **Vitamin D:** Aids calcium absorption and bone development (sunlight, fortified foods, fatty fish).
- **Iodine:** Supports thyroid function and baby's brain development (iodized salt, seafood, dairy).
- **Vitamin B12:** Crucial for energy and red blood cell formation (animal products, fortified cereals).
- **Omega-3 Fatty Acids (DHA):** Enhances baby's brain and eye development (salmon, flaxseeds, walnuts).

3. Hydration

- Increased water intake is essential for milk production (8-12 glasses/day).

Dietary Guidelines for Lactating Mothers

1. Balanced Diet

- Include all food groups: whole grains, proteins, fruits, vegetables, and healthy fats.
- Avoid excessive processed foods, sugary snacks, and high-salt items.

2. Meal Frequency

- Frequent meals and healthy snacks to meet energy demands.

3. Special Foods

- Lactogenic foods to support milk production: oats, fennel seeds, fenugreek, and almonds.

4. Safe Food Practices

- Avoid alcohol, caffeine (limit to 1-2 cups/day), and potential allergens if the baby shows sensitivity.

Common Challenges and Solutions

1. Fatigue and Lack of Time

- opt for quick, nutrient-rich snacks like nuts, boiled eggs, or smoothies.
- Meal prep in advance.

2. Low Milk Supply

- Stay hydrated and consume lactogenic foods.
- Ensure adequate calorie and protein intake.

3. Nutrient Deficiencies

- Supplement vitamins and minerals like Vitamin D, B12, and iron as prescribed.

4. Weight Management

- Gradual weight loss through balanced eating and light exercise. Avoid crash diets.

NUTRITION REQUIREMENTS DURING ADOLESCENT

Adolescence is the period between childhood and adulthood. It is a period of rapid growth after infancy and it reaches its peak between 11th to 14th years for girls and 13th to 16th years for boys. Adolescence requires more food for the following reasons:

- a. This period (teenage) is spread almost over a decade,
- b. It is characterized by rapid increase in height and weight, hormonal changes, sexual maturation and wide swings in emotion,
- c. Development of critical bone mass is essential during this period as this forms the ground for maintaining mineral integrity of the bone in later life,
- d. The pattern and proportion of various body components like body water, muscle mass, bone and fat increase during the entire childhood and adolescence to reach adult values by about 18 years.

Adolescent girls are at greater physiological stress than boys because of menstruation. Their nutritional needs are of particular importance as they have to prepare for motherhood.

NUTRITIONAL NEEDS OF ADOLESCENTS

Good nutrition is critical during the teen-age years to ensure healthy growth and development. A healthy diet must meet the changing nutritional needs of a growing teenager.

Energy

Calorie needs are influenced by activity level, basal metabolic rate, increased requirements to support pubertal growth, development and energy expenditure. The energy requirements for boys are more than that of girls.

Protein

Protein needs of adolescents are influenced by the amount of protein required for maintenance of existing lean body mass and accrual of additional lean body mass during the adolescent growth spurt. When protein intakes are consistently inadequate, reductions in linear growth, delays in sexual maturation and reduced accumulation of lean body mass may be seen.

Fat and Essential Fatty Acids

The human body requires dietary fat and essential fatty acids for normal growth and development. The intakes of total fat and saturated fat should not exceed RDA.

Calcium

Its needs during adolescence are greater than they are in either childhood or adulthood because of the dramatic increase in skeletal growth. Milk provides the greatest amount of calcium in the diets of adolescents. Ragi, green leafy vegetables, milk and milk products are excellent sources of calcium.

Iron

Iron is vital for transporting oxygen in the bloodstream and for preventing anemia. For both male and female adolescents, the need for iron increases with rapid growth and the expansion of blood volume and muscle mass. The onset of menstruation imposes additional iron needs for girls.

Zinc

Zinc is important in adolescence because of its role in growth and sexual maturation. Males who are zinc deficient experience growth failure and delayed sexual development.

Vitamins

Vitamin A is important for normal vision and plays a vital role in reproduction, growth, and immune function. The most obvious symptom of inadequate vitamin A consumption is vision impairment, especially night blindness. The low intake of fruits, vegetables and milk and dairy products by adolescents contributes to their less-than-optimal intake of vitamin A.

Vitamin E is well known for its anti-oxidant properties, which become increasingly important as body mass expands during adolescence. Fortified breakfast cereals and nuts are good sources of vitamin E. Vitamin C is involved in the synthesis of collagen and other connective tissues

Fiber

Dietary fiber is important for normal bowel function, and plays a role in the prevention of chronic diseases, such as certain cancers, coronary artery disease, and type 2 diabetes mellitus and reduces the risk of obesity. Increased intake of fruit, vegetables, and whole grains increases the fiber intake. Adolescents who skip breakfast or do not routinely consume whole grain cereals are at high risk for having an inadequate consumption of fiber.

Table 11 Balanced Diet for Adolescents (Number of portions)

Food groups	g/Portion	10-12 Years		13-15 Years		16-18 Years	
		Girls	Boys	Girls	Boys	Girls	Boys
Cereals & millets	30	8	10	11	14	11	15
Pulses	30	2	2	2	2.5	2.5	3
Milk & its products	100	5	5	5	5	5	5
Roots & tubers	100	1	1	1	1.5	2	2
Green leafy veg.	100	1	1	1	1	1	1
Other vegetables	100	2	2	2	2	2	2
Fruits	100	1	1	1	1	1	1
Sugar	5	6	6	5	4	5	6
Fat/oil (visible)	5	7	7	8	9	7	10

Source: Dietary Guidelines for Indians, National Institute of Nutrition, Hyderabad, 2011.

DIETARY PATTERNS OF ADOLESCENTS

- Adolescence is a critical growth phase characterized by rapid physical, emotional, and cognitive development.
- Proper nutrition during this period supports growth, hormonal changes, and overall health.

Key Nutritional Requirements

1. Macronutrients

- **Carbohydrates:** Primary energy source (whole grains, fruits, vegetables).
- **Proteins:** Essential for growth and muscle development (eggs, lean meat, dairy, legumes, nuts).
- **Healthy Fats:** Support brain development and hormone production (nuts, seeds, avocados, fatty fish).

2. Micronutrients

- **Calcium:** Strengthens bones and teeth (milk, cheese, yogurt, fortified alternatives).
- **Iron:** Prevents anemia and supports oxygen transport (meat, spinach, lentils, fortified cereals).
 - **Girls:** Higher iron needs due to menstruation.
- **Vitamin D:** Enhances calcium absorption and bone health (sunlight, fortified foods, fatty fish).
- **Zinc:** Supports immune function and growth (nuts, seeds, whole grains).
- **Vitamin C:** Boosts immunity and enhances iron absorption (citrus fruits, tomatoes, bell peppers).

3. Hydration

- Adequate water intake is critical for overall health and energy.

Dietary Guidelines for Adolescents

1. Balanced Diet

- Include all food groups: carbohydrates, proteins, fruits, vegetables, and fats.
- Emphasize whole, nutrient-dense foods over processed options.

2. Meal Frequency

- Regular meals and healthy snacks to meet energy demands.
- Examples of snacks: nuts, fresh fruits, boiled eggs, yogurt.

3. Portion Control

- Educate on appropriate portion sizes to prevent overeating or under-eating.

4. Limitations

- Reduce intake of sugary beverages, snacks, and fast foods.

Common Challenges and Solutions

1. Skipping Meals (Especially Breakfast)

- Offer quick and healthy breakfast options like smoothies, whole-grain toast, or boiled eggs.

2. Peer Pressure and Fast-Food Consumption

- Educate on making healthier choices when eating out.
- Include occasional treats to maintain balance.

3. Nutrient Deficiencies

- Address through fortified foods or supplements if needed.
- Encourage a variety of foods to ensure nutrient diversity.

4. Body Image Issues

- Promote positive body image and discourage fad diets.
- Focus on health rather than weight.

Special Considerations

1. Increased Nutritional Needs

- Boys may require more calories and proteins for muscle development.
- Girls may require more iron due to menstruation.

2. Physical Activity

- Nutrition should align with activity levels, especially for adolescents involved in sports.

3. Cultural and Economic Factors

- Tailor dietary advice to local food availability and cultural preferences.

NUTRITIONAL NEEDS OF ADULTS

When an individual reaches adulthood, body growth especially in terms of height and body status stop to a certain extent, but tissue breakdown and repair of body tissues continue even among adults. Therefore, adequate amount of essential nutrients needs to be provided for maintenance of physical and mental health in adults.

Energy-Kilocalories

There is a gradual loss of functioning body cells and reduced physical activity so adults generally require less energy intake as they grow older. The basic fuels required to supply these energy needs are primarily carbohydrates with moderate fat.

Protein

The RDA for an adult necessity a protein intake of 0.8g/kg of body weight making the total protein. This amount of protein provides about 13-15% of the total calorie.

Carbohydrates

About 50-60% of the total diet calories should come from carbohydrate foods, with the majority being mostly complex carbohydrates such as starches. Easily absorbed sugars may also be used for immediate energy.

Fat

It provides a back-up energy source. Sufficient fat makes food taste better, aids appetite and provides needed kcal to pre-vent excessive weight loss.

Calcium and Phosphorus

In adults, calcium is required for replacing calcium lost from body through urine, feces, sweat and bile. Of the dietary calcium only 20-30% is absorbed and this is facilitated by vitamin D. A desirable intake of phosphorus is recommended. The elemental Ca :P ratio in the diet should be maintained at 1:1.

Iron

The loss of iron through sweat, gastrointestinal tract and urine is estimated to be 14 mg/kg body weight. Apart from this woman have additional loss due to menstruation. Thus, the iron requirements for women are more than men.

Vitamins

Studies have revealed that 600 mg of retinol daily would be sufficient to maintain a normal serum vitamin A level. The requirement for B Vitamins is based on calorie intake. Requirement of folic acid among Indians is 200µg. A daily intake of 20 mg vitamin C is sufficient to maintain ascorbic acid status. Since 50 percent vitamin C is lost during cooking 40 mg of vitamin C per day is recommended.

Table 12 Balanced Diet for Adults - Sedentary/ Moderate/ Heavy Activity

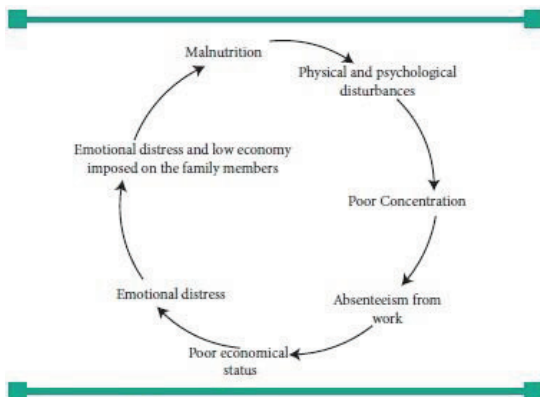
S. No	Food Groups	Portion g/Day	Activity					
			Sedentary		Moderate		Heavy	
			Man	Women	Man	Women	Man	Women
			Number Of Portions					
1.	Cereals And Millets	30	12.5	9	15	11	20	16
2.	Pules	30	2.5	2	3	2.5	4	3
3.	Milk	100ml	3	3	3	3	3	3
4.	Roots And Tubers	100	2	2	2	2	2	2
5.	Green Leafy Vegetables	100	1	1	1	1	1	1
6.	Other Vegetables	100	2	2	2	2	2	2
7.	Fruits	100	1	1	1	1	1	1
8.	Sugar	5	4	4	6	6	11	9
9.	Fats/ Oil Visible	5	5	4	6	5	8	6

Source: Dietary Guidelines for Indians A Manual, NIN, Hyderabad (2011).

Nutritional Problems for an Adult

- ☐ Osteoporosis
- ☐ Anemia
- ☐ Chronic Disease
- ☐ Diabetes Mellitus
- ☐ Underweight
- ☐ Coronary Heart Disease (CHD)
- ☐ Poor Nutritional Status

Many of the health problems of older adults are not only due to general aging but also due to states of malnutrition.



▲ Fig. 6.4 Malnutrition

DIETARY PATTERNS OF ADULTS

- Adulthood is a phase where maintaining a balanced diet is crucial for sustaining energy levels, preventing chronic diseases, and supporting overall health.
- Nutritional needs vary based on age, activity level, gender, and health status.

Key Nutritional Requirements

1. Macronutrients

- **Carbohydrates:** Primary energy source (whole grains, fruits, vegetables).
- **Proteins:** Supports muscle maintenance and repair (lean meats, fish, eggs, legumes, dairy).
- **Fats:** Healthy fats for brain and heart health (avocado, nuts, seeds, olive oil, fatty fish).

2. Micronutrients

- **Calcium & Vitamin D:** Bone health and muscle function (milk, yogurt, fortified foods, sunlight).
- **Iron:** Prevents anemia (meat, spinach, lentils, fortified cereals).
- **Vitamin B12:** Supports nerve function and red blood cell formation (animal products, fortified cereals).
- **Potassium:** Regulates blood pressure (bananas, potatoes, oranges, spinach).
- **Antioxidants (Vitamins C & E):** Protects cells from damage (citrus fruits, nuts, seeds, leafy greens).

3. Water and Hydration

- Drink 8-10 glasses of water daily; adjust based on climate and activity levels.

Dietary Guidelines for Adults

1. Balanced Diet

- Emphasize fruits, vegetables, whole grains, lean proteins, and healthy fats.
- Follow the "plate method": Half the plate vegetables and fruits, a quarter whole grains, and a quarter lean protein.

2. Portion Control

- Avoid overeating; adjust portions based on physical activity and caloric needs.

3. Limitations

- Reduce intake of saturated fats, trans fats, added sugars, and sodium.
- Limit processed foods and sugary drinks.

4. Meal Timing

- Regular meals and healthy snacks to stabilize blood sugar and energy levels.

Common Challenges and Solutions

1. Busy Lifestyles

- Opt for quick, healthy meals like salads, whole-grain wraps, and stir-fries.
- Plan and prep meals in advance.

2. Weight Management

- Focus on nutrient-dense foods rather than empty calories.
- Include physical activity to complement dietary efforts.

3. Chronic Conditions

- Tailor diets for specific conditions (e.g., DASH diet for hypertension, low-GI diet for diabetes).
- Consult healthcare providers for personalized plans.

4. Nutrient Deficiencies

- Ensure variety in the diet to meet nutritional needs.
- Consider fortified foods or supplements if required.

Special Consideration

1. Gender-Specific Needs

- Women: Increased calcium and iron needs.
- Men: Focus on heart health with omega-3 fatty acids and fiber.

2. Age-Related Changes

- Metabolism slows with age, necessitating adjustments in caloric intake.
- Emphasize foods that support muscle mass and bone density.

3. Cultural and Economic Factors

- Adapt meal plans to cultural preferences and budget constraints.

NUTRITIONAL NEEDS AND CHALLENGES DURING OLD AGE

Individuals above the age of 60 years constitute the elderly. Health and well-being of the elderly is given more importance and has paved way for a specific field of study called “Geriatric Nutrition”.

Aging

Aging is an irreversible biochemical change that occurs throughout an individual’s life cycle and continues until death. In old age, the nutritional status is determined by the state of nutrition of an individual’s cell. Conditions like dietary deficiency, improper digestion and absorption, insufficient distribution of nutrients, accumulation of excess waste products lead to poor nourishment of cells.

Table 14 A Balanced Vegetarian Diet For Pregnant and Lactating Mother Doing Sedentary Work

S.No	Food group	Pregnant mother Quantity (g)	Lactating mother Quantity (g)
1	Cereals millets	300	330
2	Pulses	60	90
3	Milk(ml)	500	500
4	Roots and tubers	100	100
5	Green leafy vegetable	150	150
6	Other vegetable	100	100
7	Fruit	200	200
8	Sugar	20	20
9	Fats and oil(visible)	30	30

Source: Dietary Guidelines for Indians - A manual, National Institute of Nutrition, ICMR, Hyderabad, India, (1999)

With increasing age, cell functioning is slowed down and their response to hormones and synthesis of enzymes decreases. These changes are associated with a decrease in the number of functioning cells. Since the cells are interdependent, if one cell dies, others of the same or different organ lose their ability to function resulting in malfunctioning of the organ. This in turn gradually affects the total body functioning leading to death.

NUTRITIONAL NEEDS DURING OLD AGE

Energy

The calorie intake should be adjusted to maintain weight. The energy requirement decreases due to the following reasons:

- Decreased physical activity
- Reduction in lean body mass and increase in adipose tissue
- Decrease in resting metabolic rate by 15-20 percent due to changes in body composition and physical inactivity

Protein

The protein requirements do not change during old age. It is the same as adults i.e. 1g/kg of body weight.

Carbohydrate

Since the calorie requirement is reduced, the carbohydrate content should also be proportionately reduced. Due to impaired glucose tolerance and gastrointestinal disturbances like constipation, emphasis should be on taking complex carbohydrates.

Fat

The fat requirements are also reduced, corresponding to the energy requirements. The intake of saturated fats and cholesterol should be less and unsaturated fat should be used in cooking.

Minerals

Calcium needs increase during old age due to increasing mobilization of calcium from bones and incidence of osteoporosis. During old age 1000mg of calcium is recommended per day because of the following reasons.

- Limited availability of calcium from foods
- To compensate age related bone loss and to improve calcium balance
- To prevent fractures and tooth decay
- To compensate decreased efficiency of calcium absorption

Aging does not affect iron needs. Hence iron needs are same as that for adults. Mild anemia may affect the health of old people due to inefficient circulation. Therefore, iron intake should be adequate during old age.

Vitamins

Vitamin A requirements remain the same i.e., 600 µg of retinol. Inadequate exposure to sunshine may affect vitamin D levels. The antioxidant vitamins such as vitamin E, carotenoids and vitamin C have been identified to promote health of the elderly. Vitamin B6 requirements are also increased due to gastritis which interferes with absorption. Besides these various nutrients, water should be consumed in plenty as such or as fluids like buttermilk, fruit juice and soups. Intake of sufficient fluids reduces the load on kidneys and relieves from constipation by stimulating peristalsis.

Table 15 Sample Menu

Time	Food Items
Early morning	Milk/tea/coffee
Morning	Idli/dosa Tomato chutney or vegetable sambar Fruit-1 slice
Mid-morning	Butter milk/soups/boiled eggs/ fruit salads
Lunch	Rice, sambhar, greens kootu, vegetable poriyal
Evening	Fruit yogurt/boiled sundal/custard
Dinner	Idiyappam/ idly/ dosa,vegetable stew/dhal,fruit.
Bedtime	Milk

NUTRITION RELATED PROBLEMS OF ELDERLY

The elderly are at risk of poor nutrition due to economic pressure, poor dentition, aging tissues and inadequate diet, which may be compounded with the incidence of chronic disease. The commonly prevalent nutrition related problems among the aged include:

- Osteoporosis
- Obesity
- Anemia
- Malnutrition
- Constipation
- Diabetes Mellitus
- Cardiovascular disease

Iron deficiency anemia is a serious public health problem throughout Central America, Like in India, Rice a staple food in Central America. It is typically polished and rarely iron-fortified.

DIETARY PATTERNS OF OLD AGE

- Aging brings physiological, metabolic, and lifestyle changes that affect dietary requirements.
- A balanced diet helps manage chronic conditions, maintain physical and mental health, and improve the quality of life.

Key Nutritional Requirements

1. Macronutrients

- **Carbohydrates:** Prefer complex carbs for sustained energy and blood sugar control (whole grains, oats, legumes).
- **Proteins:** Preserve muscle mass and prevent sarcopenia (lean meats, fish, eggs, dairy, tofu).

- **Healthy Fats:** Support brain and heart health (olive oil, nuts, seeds, fatty fishlike salmon).
- 2. **Micronutrients**
 - **Calcium & Vitamin D:** Essential for bone health and preventing osteoporosis (milk, fortified plant-based milks, sunlight exposure).
 - **Vitamin B12:** Prevents anemia and supports neurological health (meat, fish, fortified cereals).
 - **Iron:** Maintains red blood cell production (lean meats, spinach, lentils).
 - **Potassium:** Regulates blood pressure and heart health (bananas, potatoes, beans).
 - **Fiber:** Supports digestive health and prevents constipation (whole grains, fruits, vegetables).
 - **Antioxidants (Vitamins A, C, and E):** Protect cells and support immune function (citrus fruits, carrots, nuts).
- 3. **Hydration**
 - Older adults are more prone to dehydration; aim for 6-8 glasses of water daily.

Dietary Guidelines for Older Adults

- 1. **Balanced Diet**
 - Include a variety of nutrient-dense foods: whole grains, vegetables, lean proteins, and healthy fats.
 - Focus on smaller portions with high nutrient density.
- 2. **Meal Frequency**
 - Eat smaller, frequent meals to maintain energy levels and prevent fatigue.
- 3. **Limitations**
 - Reduce intake of added sugars, saturated fats, and salt to manage chronic conditions like hypertension and diabetes.
- 4. **Special Foods**
 - Include easily digestible and soft-textured foods for those with dental issues.

Common Challenges and Solutions

- 1. **Decreased Appetite**
 - Enhance flavor with herbs and spices instead of salt.
 - Offer nutrient-dense snacks like yogurt, nuts, and fruit.
- 2. **Difficulty Chewing or Swallowing**
 - Choose soft or pureed foods like soups, stews, mashed potatoes, or smoothies.
- 3. **Digestive Issues**
 - Increase fiber intake gradually and stay hydrated to prevent constipation.

4. Nutrient Deficiencies

- Use fortified foods or supplements for calcium, Vitamin D, and B12 as advised by a healthcare provider.

Special Considerations

1. Chronic Conditions

- Tailor diets to manage conditions like diabetes (low-GI foods), hypertension (DASH diet), or heart disease (low-fat, high-fiber diet).

2. Physical Activity

- Encourage light activities like walking or yoga to support metabolism and overall health.

3. Economic and Social Factors

- Plan meals that are affordable and easy to prepare.
- Consider community programs or meal delivery services for support.

4. Hydration Awareness

- Monitor water intake as thirst perception decreases with age.

DIETARY MODIFICATION DURING OLD AGE

Dietary modifications	Reasons
Foods must be soft and easily chewable	Problems of dentition -fallen teeth or dentures
Foods should be easily digestible	Decreased production of digestive enzymes
Restricted fat in diet, inclusion of Poly unsaturated fatty acid (PUFA)	Susceptible to heart diseases
Foods rich in fibre should be given	To prevent constipation, reduce cholesterol levels. Also to prevent colon cancer
Coffee, tea, cold beverages should be restricted	May result in insomnia
Foods rich in calcium such as milk should be given	To compensate bone loss and prevent osteoporosis
Green leafy vegetables can be given liberally	Source of nutrients like: iron riboflavin, folic acid, vitamin c, antioxidants, carotene, and fibre
Familiar foods should be included (others difficult to digest)	Unfamiliar or changes in the food pattern- may lead to Psychological problems-depression and anxiety
New foods are difficult to accept	
Clear soup can be given at the beginning of the meal	Aids digestion
Small and frequent meals should be given instead of 3 heavy meals	Favours complete digestion, prevents distress
Glass of milk can be given before sleep	Induces sleep
Heavy meal-lunch	Sleep is less likely to be disturbed
Light meal-dinner	
Too many sweets with lots of fats and sugar should be avoided	Too much of sugar may cause fermentation, discomfort-due to indigestion and causes tooth ache and increases cholesterol level. May lead to obesity
Plenty of fluids	To prevent constipation and dehydration

UNIT II-MACRO NUTRIENTS AND MICRONUTRIENTS

Macronutrients. Carbohydrates, Lipids, Proteins, their dietary source and role. Micronutrients. Vitamins- Water-soluble and Fat-soluble vitamins- their sources and importance. Important minerals viz., Iron, Calcium, Phosphorus, Iodine, Sodium their biological functions.

Macronutrients (carbohydrates, proteins, and lipids) are needed in large quantities by the body and provide energy and structural components. Micronutrients (vitamins and minerals) are needed in smaller amounts and play crucial roles in metabolism and various physiological functions.

CARBOHYDRATES:

Carbohydrates are the main source of energy for our body. There are three major categories of carbohydrates, viz. starch, sugar and cellulose. Our body breaks down sugar and starch into glucose to provide energy. Cellulose does not have any nutritive value per se, but is a major contributor of dietary fibres.

Excess carbohydrates are converted into fat by the liver and stored in the adipose tissues in our body. Carbohydrates are of two types – simple and complex. Simple carbohydrates are quick energy foods, e.g. sugar. Sources of simple carbohydrates are natural fruits, milk and milk products, and vegetables including potatoes and carrots. Complex carbohydrates are better sources of energy than sugar since these are released slowly. Sources of complex carbohydrates are breads, cereals (rice, wheat, bajra, corn, barley, ragi, etc.).

DIETARY SOURCES:

- **Starchy foods:** Potatoes, bread, rice, pasta, and cereals are major sources of carbohydrates.
- **Fruits and Vegetables:** Provide natural sugars, fiber, and complex carbohydrates.
- **Dairy Products:** Milk and yogurt contain carbohydrates, mainly in the form of lactose.
- **Legumes and Seeds:** Beans, lentils, and seeds are good sources of complex carbohydrates and fiber.
- **Simple Carbohydrates:** Found in fruits, fruit juices, vegetables, honey, and dairy, as well as in processed sugars like table sugar, brown sugar, and high fructose corn syrup.

ROLE

- **Energy Source:**

Carbohydrates are the body's main fuel source, providing energy for daily activities and physical exertion.

- **Sparing Proteins:**

When sufficient carbohydrates are available, the body can use them for energy instead of proteins, conserving protein for other vital functions.

- **Fat Metabolism:**

Carbohydrates play a role in the metabolism of fats, helping the body utilize them for energy.

- **Digestive Health:**

Dietary fibre, a type of carbohydrate, promotes digestive health by aiding in bowel movements and preventing constipation.

- **Blood Glucose Regulation:**

Carbohydrates influence blood sugar levels, and consuming complex carbohydrates helps regulate blood glucose and prevent rapid spikes.

- **Brain Function:**

Carbohydrates are the primary fuel source for the brain, supporting cognitive function and nerve tissue regulation.

- **Structural Components:**

Carbohydrates are also involved in forming structural components like cell walls in plants and microorganisms.

- **Surface Antigens and Receptor Molecules:**

Carbohydrates can associate with lipids and proteins to form surface antigens and receptor molecules.

PROTEINS

Proteins are called building blocks of our body. About 20 per cent of our body weight comes from proteins. Proteins are essential for growth and repair of muscle and other body tissues. Proteins are made of amino acids that contain the elements carbon, hydrogen, oxygen and nitrogen.

DIETARY SOURCES OF PROTEIN

- **Animal Sources:**

Meat (beef, chicken, pork), fish (salmon, tuna), dairy (milk, yogurt, cheese), and eggs are excellent sources of protein and contain all essential amino acids (complete proteins).

- **Plant Sources:**

Legumes (beans, lentils, chickpeas), nuts (almonds, walnuts), seeds (chia, flax), soy products (tofu, tempeh), and grains (quinoa, oats) are also good sources of protein, though some may be incomplete (lacking one or more essential amino acids).

- **Complete Proteins:**

Animal sources and some plant sources like soy, quinoa, and amaranth contain all nine essential amino acids.

- **Incomplete Proteins:**

Most plant-based proteins are considered incomplete, as they are missing at least one essential amino acid.

ROLE OF PROTEIN IN THE BODY:

- **Tissue Growth and Repair:**

Protein is crucial for building and repairing muscles, bones, skin, and other tissues.

- **Enzyme and Hormone Production:**

Proteins are essential components of enzymes that catalyze biochemical reactions and hormones that regulate various bodily functions.

- **Immune Function:**

Proteins are part of antibodies that help the body fight infections.

- **Energy Source:**

While not the primary source of energy, protein can be used for energy when the body doesn't have enough carbohydrates or fats.

- **Transport and Storage:**

Some proteins transport oxygen and nutrients throughout the body, while others store nutrients like iron.

- **Fluid and Electrolyte Balance:**

Proteins help maintain the proper balance of fluids and electrolytes in the body.

FATS

Fats and oils are concentrated sources of energy. Fats are substances that our body stores for future use. These are classified as simple lipids (e.g. triglycerides), compound lipids (e.g. phospholipids), and derived lipids (e.g. cholesterol). Fats may be obtained from animal or vegetable sources. The major sources of animal fats are ghee, butter, milk, cheese, eggs, fish and meat. The sources of vegetable oils are groundnut, mustard, sesame, coconut, etc. Excess carbohydrates are converted into fats. Fast food items like pizza, samosas, burgers are rich sources of fat. Therefore, children who consume only fast-food items tend to become obese.

DIETARY SOURCES:

- **Animal-based:** Meats, dairy products (butter, milk, cheese), and fish.
- **Plant-based:** Nuts, seeds, fruits (avocado), and vegetable oils (olive, canola, sunflower).

ROLE:

- **Energy Source:** Fats provide a concentrated source of energy, delivering approximately 9 calories per gram.
- **Nutrient Absorption:** Fats help the body absorb fat-soluble vitamins (A, D, E, K).
- **Cellular Structure:** Fats are a major component of cell membranes.
- **Hormone Production:** Fats are involved in the synthesis of essential hormones.
- **Insulation and Protection:** Fats help insulate the body and cushion vital organs.
- **Satiety:** Fats contribute to feelings of fullness after a meal.
- **Essential Fatty Acids:** Fats provide essential fatty acids (e.g., omega-3 and omega-6), which the body cannot synthesize itself and are vital for various functions like brain development and inflammation control.

VITAMINS

Vitamins are organic substances that we require in adequate quantities for good health. Our body, however, cannot synthesize them. Therefore, we need to consume their natural sources such as fruits and vegetables. Vitamins are divided into two groups—fat soluble vitamins (Vitamins A, D, E, and K) and water-soluble vitamins (Vitamins of B group and Vitamin C).

Fat-soluble vitamin dissolve in fat before they are absorbed in the bloodstream to carry out their functions. Each vitamin has a specific function and its deficiency leads to a particular deficiency disease. Excess of these vitamins are stored in the liver. Because they are stored, they are not needed every day in the diet. By contrast, water-soluble vitamins dissolve in water and are not stored. They are eliminated in urine.

VITAMINS	DIETARY SOURCES OF VITAMINS
WATER SOLUBLE VITAMINSs	
Vitamin B ₁	Pulses, meats, yeasts, cereals, and sprouted beans
Vitamin B ₂	Yoghourts, Pulses, green vegetables, milks and yeasts
Vitamin B ₃	Fishes, meats, cereals, grains, pulses, yeasts and livers
Vitamin B ₅	Eggs, yeasts and most of the foods
Vitamin B ₆	All plant and animal tissues
Vitamin B ₇	Yeasts, livers and kidneys
Vitamin B ₉	Fishes, livers and vegetables. It is also produced by bacteria in colon
Vitamin B ₁₂	Milks, kidneys, meats, eggs and fishes. Bacteria in the colon of humans and in the digestive tract of cattle also produce this vitamin
Vitamin C	Green vegetables, Citrus fruits like oranges and lemons
FAT SOLUBLE VITAMINS	
Vitamin A	Cod liver oils, carrots, maize, spinaches, butters, and milks
Vitamin D	Cod liver oils, by the action of sunlight on the skin, shark liver oils and eggs
Vitamin E	Brown flours, vegetable oils, green vegetables, germinated wheats and seeds
Vitamin K	Wheat germs, cabbages and corianders. It is also synthesised by the bacteria of the large intestine

VITAMIN A

DIETARY SOURCES

1. Only in fish liver oil and foods of animal origin such as liver, eggs, milk and fatty fish.
2. Plant foods contain only carotenoids (the precursor of vitamin A) and include green leafy vegetables, tomato, carrot, mango and other yellow-colored fruits.

FUNCTIONS

1. Maintenance of normal vision.
2. Building and growth of skeletal cells and provides immunity to the body.
3. Dietary retinoids, especially carotenoid compounds have been found to suppress carcinogenesis (development of cancer).

VITAMIN D

DIETARY SOURCES

1. Vitamin D₂ occurs in small amounts in fish liver oils.
2. Vitamin D₃ is widely distributed in eggs, milk, butter and cheese but large amounts occur only in fish liver oils.

FUNCTIONS

1. Essential for bone formation.
2. Promotes absorption of calcium and phosphorous and deposition in bone.
3. Involved in the formation of teeth.

VITAMIN E

DIETARY SOURCES

1. Delta tocopherols were isolated from soybean oil as the richest natural sources
2. Cereal germ oils like wheat germ oil and corn germ oils, soybean oil.
3. Cottonseed oil contains alpha, beta, and gamma tocopherols.

FUNCTIONS

Essential for normal reproduction in several species of animals and also in human beings.

VITAMIN K

1. Occurs widely in plant foods, especially in leafy vegetables like spinach, cabbage, alfa-alfa.
2. Almost 50% of the vitamin requirement is derived from intestinal micro-organisms.

FUNCTIONS

Essential for blood clotting by increasing the prothrombin levels in blood

WATER SOLUBLE VITAMINS

VITAMIN B1 (THIAMINE)

Whole cereals, pulses (legumes), oilseeds, eggs, pork and nuts.

FUNCTIONS

1. Functions in carbohydrate metabolism.
2. Free thiamine is readily absorbable by small intestine.
3. Necessary for Nerve function, appetite and normal Digestion.

VITAMIN B2 (RIBOFLAVIN)

Liver, dried yeast, white bread, egg powder, whole and skim milk excellent sources. Milk, cheese, eggs, whole grain and green leafy vegetables are also good sources.

FUNCTIONS

Concerned in the regulatory function of insulin. The retina riboflavin, which is converted by light to a compound involved in stimulation of the optic nerve. Forms a part of enzyme systems involved in the metabolism of carbohydrates, fats and proteins.

VITAMIN B5 (PANTOTHENIC ACID)

Widely distributed in foods. Dried yeasts, liver, rice polishing, wheat germs, fleshy foods, eggs, fish etc. are good sources.

FUNCTIONS

1. Important role in the metabolism of Co-enzyme A
2. Indirectly it has a role in the utilization of carbohydrates and fats

VITAMIN B6 (PYRIDOXINE HYDROCHLORIDE)

Both plant and animal foods. Dried yeast, rice polishing, wheat germ and liver are excellent sources. Whole cereals, legumes, oil seeds, nuts, egg, milk, meat and fish and leafy vegetables are good sources

FUNCTIONS

1. Essential for growth of infants
2. Important roles in amino acid and lipid metabolism
3. Pyridoxal phosphate acts as a coenzyme for a number of enzyme systems, removes carbon dioxide from the acid groups of certain amino acids and transfer amine groups from one compound to another
4. Pyridoxal phosphate helps in transamination reactions, porphyrin synthesis etc.

NIACIN (NICOTINIC ACID)

Yeast, liver, meat, poultry, whole grains, fresh pork sources of niacin. Good proteins like milk protein are associated with niacin because tryptophan, an amino acid, present in the proteins is converted into niacin in the body.

FUNCTION

It is essential for the normal functioning of the skin, intestinal tract and the nervous system

BIOTIN

Peanuts, chocolates, egg yolk, liver, kidney, peas, cauliflower, dry yeast, milk products, cereals etc. are good sources. Royal jelly from honeybee is the richest source of biotin. Produced by intestinal bacteria.

FUNCTION

1. Essential for the activity of many enzymes systems
2. Helps in maintaining the skin structure and is necessary for normal gestation and lactation in animals

3. Required for fatty acid metabolism

FOLIC ACID (PTEROYLGLUTAMIC ACID)

Dried yeast, green leafy vegetables, dry beans, cabbage, soybean, yeast, kidney and liver

FUNCTION

- 1.. Essential for reproduction in animals
2. Also helps in the hair growth and health of skin.

VITAMIN B12 (CYANOCOBALAMIN)

Present only in foods of animal origin. Kidney and liver, egg, cheese, milk, fish etc. are good sources

FUNCTION

Takes part in many enzymatic reactions, is essential for the absorption of calcium and phosphorus.

IRON

The total content of iron in an adult body is 3-5 g. About 70% of this occurs in the erythrocytes of blood as a constituent of hemoglobin. At least 5% of body iron is present in myoglobin of muscle. Haem is the most predominant iron-containing substance. Iron is mostly found in the foods in ferric form (Fe^{3+}), bound to proteins or organic acids. In the acid medium provided by gastric HCl, the Fe^{3+} is released from foods. Reducing substances such as ascorbic acid (vitamin C) and cysteine convert ferric iron (Fe^{3+}) to ferrous form (Fe^{2+}). Iron in the ferrous form is soluble and readily absorbed. It is stored in body in the form of ferritin.

SOURCES: Rich sources, Leafy vegetables (spinach, silver beet, broccoli), pulses, apples, organ meats (liver, heart, kidney), fish, molasses, dry fruits, nuts, tofu and Jaggery. Other sources Milk, wheat, polished rice, oats, egg etc. Cooking in iron utensils will improve the iron content of the diet.

RECOMMENDED DIETARY REQUIREMENTS:

- Adult man — 10 mg/day
- Menstruating woman — 18 mg/day
- Pregnant and lactating woman — 40 mg/day

FUNCTIONS:

a) Oxygen transport: Iron is a major constituent of a red-colored compound called hemoglobin present in the blood. Iron is present in the haem part of hemoglobin. Hemoglobin is necessary for transport of oxygen to various parts of the body. Hemoglobin carries oxygen from the lungs to the tissues and in turn helps in carrying carbon dioxide from the tissues to the lungs.

b) Provision of oxygen for muscle contraction: Iron is also present in the muscle in the form of myoglobin. Myoglobin has the capacity to store oxygen. This oxygen is used for muscle contraction and for other immediate needs of the muscle cells.

c) Promotion of oxidation within cells: Iron facilitates the complete oxidation of carbohydrates, fats and proteins within the cell. This, of course, would result in the release of the energy locked up in these molecules. The role of iron in oxygen transport and release of energy.

d) Iron plays an important role in maintenance of specific brain functions like immediate memory, capacity to learn and attention span.

e) Iron forms a vital component of certain enzymes and substances that aid in metabolism.

f) Iron has protective functions as well. Like vitamin A, iron too helps in preventing infections.

CALCIUM

Calcium is the most abundant among the minerals in the body. The total content of calcium in an adult man is about 1 to 1.5 kg. As much as 99% of it is present in the bones and teeth. A small fraction (1%) of the calcium, found outside the skeletal tissue. Absorption is taking place from the first and second part of duodenum. Calcium is absorbed against a concentration gradient and requires energy. Absorption requires a carrier protein, helped by calcium-dependent ATPase.

SOURCES:

Milk and milk products are the good source of calcium (cow's milk, yogurt, cheese). Egg, fish and green vegetables, fruits (Papaya, orange, kiwifruit) are also medium sources for calcium. Other than these, cereals (wheat, rice), dry fruits (almond), beans and some pulses also contain the calcium.

DIETARY REQUIREMENTS

Daily requirement of calcium by age

1-3 years of age- 700 mg/day

4-8 years of age -1000 mg/day

9-18 years of age- 1300 mg/day

19-50 years of age -1000 mg/day

50 years of age -1200 mg/day

FUNCTIONS

1. Development of bones and teeth— Calcium, along with phosphate, is required for the formation (of hydroxyapatite) and physical strength of skeletal tissue. Bone is regarded as a mineralized connective tissue. Bones which are in a dynamic state serve as reservoir of Ca. Osteoblasts are responsible for bone formation while osteoclasts result in demineralization.

2. Muscle contraction— Ca^{2+} interacts with troponin C to trigger muscle contraction. Calcium also activates ATPase, increases the interaction between actin and myosin.

3. Nerve transmission— It is necessary for the nerve transmission by activating the axonal vesicle in the neuron.

Membrane integrity and permeability— Ca^{2+} influences the membrane structure and transport of water and several ions across it.

4. Activation of enzymes— Ca^{2+} is needed for the direct activation of enzymes such as lipase (pancreatic), ATPase and succinate dehydrogenase.

5. Release of hormone and Secretory processes— Ca^{2+} regulates microfilament and microtubule mediated processes such as endocytosis, exocytosis and cell motility and also helps in the release of certain hormone like insulin, PTH, calcitonin.

DEFICIENCY:

Hypercalcemia, Hypocalcemia, rickets and osteoporosis.

PHOSPHORUS

An adult body contains about 1 kg phosphate and it is an intracellular component of cells. Most of it (about 80%) occurs in combination with Ca^{2+} in the bones and teeth. About 10% of body phosphorus is found in muscles and blood in association with proteins, carbohydrates and lipids. Calcitriol promotes phosphate uptake along with calcium and acidity favors while phytate decrease phosphate uptake by intestinal cells.

SOURCES:

Best sources—milk and milk products, eggs, chicken meat, fish oil, seeds (sunflower, pumpkin), beans. Other sources— banana, vegetable oils, almond, nuts, cereals etc. The ratio of calcium and phosphorus is required in 1:1 ratio means (generally 800 mg/day). But in infants its ratio about 2:1. Calcium and phosphate are distributed in the majority of natural foods in 1: 1 ratio.

FUNCTIONS

1. Phosphorus is essential for the development of bones and teeth.

2. It plays a central role for the formation and utilization of high-energy phosphate compounds e.g., ATP, GTP, creatine phosphate etc.

3. Phosphorus is required for the formation of phospholipids, phosphoproteins and nucleic acids (DNA and RNA).

4. It is an essential component of several nucleotide coenzymes e.g., NAD⁺, NADP⁺, pyridoxal phosphate, ADP, AMP.
5. Phosphate buffer system is important for the maintenance of pH in the blood (around 7.4) as well as in the cells.

IODINE

Iodine is an essential nutrient crucial for thyroid hormone production, which regulates metabolism, growth, and development. Dietary requirements vary by age, with pregnant and breastfeeding women needing more. Good sources include seafood, iodized salt, and dairy products.

SOURCES

- 1. Seafood:** Fish-like cod, tuna, and salmon are excellent sources of iodine.
- 2. Iodized Salt:** Salt fortified with iodine is a common source.
- 3. Dairy Products:** Milk and yogurt can provide iodine.
- 4. Seaweed:** Some types of seaweed, like kelp, are rich in iodine.
- 5. Eggs:** Eggs are a natural source of iodine.
- 6. Bread:** In some countries, like Australia, bread is made with iodized salt.

DIETARY REQUIREMENTS

- 1. Infants (0-12 months):** Approximately 110 micrograms of iodine per day.
- 2. Children (1-10 years):** Approximately 90 micrograms of iodine per day.
- 3. Adolescents (11-18 years):** Approximately 150 micrograms of iodine per day.
- 4. Adults (19 years and older):** 150 micrograms of iodine per day.
- 5. Pregnant Women:** 220 micrograms of iodine per day.
- 6. Breastfeeding Women:** 290 micrograms of iodine per day.

FUNCTIONS

1. Thyroid Hormone Synthesis:

Iodine is a key component of thyroid hormones (T3 and T4).

2. Metabolism Regulation:

Thyroid hormones regulate metabolic rate, influencing energy use and body temperature.

3. Growth and Development:

Iodine is essential for proper bone and brain development during pregnancy and infancy.

4. Brain and Nervous System Development:

Iodine is critical for foetal brain development and is especially important for babies and young children.

SODIUM

Sodium is the chief cation of the extracellular fluid. About 50% of body sodium is present in the bones, 40% in the extracellular fluid and the remaining (10%) in the soft tissues. Sodium is readily absorbed in the gastro intestinal tract.

SOURCES:

Common salt (NaCl) used in the cooking medium is the major source of sodium. Best sources— bread, whole grains, leafy vegetables, nuts, eggs and milk, sea fish, green vegetables. Other sources— fruits (Apple, mango, banana), tomato, cucumber.

RECOMMENDED DIETARY REQUIREMENTS:

For normal individuals, the requirement of sodium is about 5-10 g/day. For patients of hypertension, around 1 g/day is recommended.

FUNCTIONS:

- In association with chloride and bicarbonate, sodium regulates the body's acid base balance or homeostatic.
- It is necessary for the normal muscle irritability and cell permeability. Sodium is involved in the intestinal absorption of glucose, galactose and amino acids.
- It is necessary for initiating and maintaining heartbeat.

DEFICIENCY DISEASES:

1. Hyponatremia: This is a condition in which the serum sodium level falls below the normal. Hyponatremia may occur due to diarrhea, vomiting, chronic renal diseases, adrenocortical insufficiency (Addison's disease).
2. Hypernatremia: This condition is characterized by an elevation in the serum sodium level. The symptoms include increase in blood volume and blood pressure. It may occur due to hyperactivity of adrenal cortex (Cushing's syndrome), prolonged administration of cortisone, ACTH and/or sex hormones.

UNIT III-MALNUTRITION AND NUTRIENT DEFICIENCY DISEASES

Definition and concept of health: Common nutritional deficiency diseases- Protein Malnutrition (e.g., Kwashiorkor and Marasmus), Vitamin A deficiency, Iron deficiency and Iodine deficiency disorders- their symptoms, treatment, Role of government and NGO organization in prevention diseases.

HEALTH

WHO Definition

The World Health Organization (WHO) defines health as a state of complete physical, mental, and social well-being, not merely the absence of disease or infirmity.

CONCEPTS OF HEALTH

Health is perceived in different ways giving rise to various concepts of health. Health has evolved over the centuries as a concept from an individual concern to a worldwide social goal.

BIOMEDICAL CONCEPT

Health means “absence of disease.” It was felt that human body is a machine and disease one of the doctors’ tasks was to repair the machine.

Developments in medical and social sciences led to is an outcome of the breakdown of the machine, and the conclusion that the biomedical concept of health was inadequate.

ECOLOGICAL CONCEPT

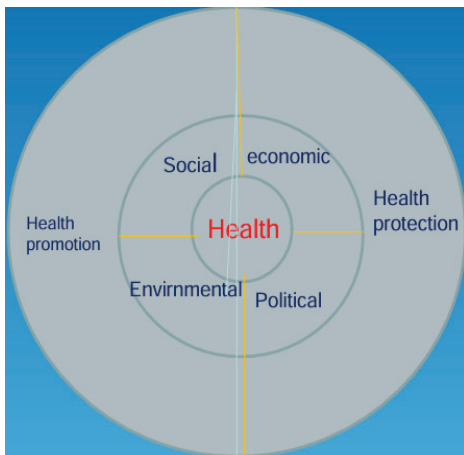
Ecologists viewed health as a dynamic equilibrium between man and his environment, and disease – a maladjustment of the human organism to environment.

PSYCHOSOCIAL CONCEPT

Advances in social sciences showed that health is not only a biomedical phenomenon, but one which is influenced by social, psychological, cultural, economic and political factors of the people concerned. Thus, health is both a biological and social phenomenon.

HOLISTIC CONCEPT

Holistic concept recognizes the strength of social, economic, political and environmental influences on health. It has been variously described as multidimensional process involving the wellbeing of the person as a whole the emphasis is on the promotion and protection of health.



PROTEIN MALNUTRITION(PEM)

PEM can be defined as a range of pathological conditions arising from a deficiency of protein and energy, and is commonly associated with infections. Protein energy malnutrition (PEM) is widely prevalent among young children (0-6 years) but is also observed as starvation in adolescents and adults, mostly lactating women, especially during periods of famine or other emergencies. PEM has serious consequences for the health of individuals particularly children and can even result in death. PEM, clinically, is characterized mainly by two forms: a) Kwashiorkor b) Marasmus

KWASHIORKOR

Kwashiorkor is distinguished by the presence of oedema (excessive accumulation of fluid in the inter-cellular spaces of tissues), low body weight for age (growth failure) and mental changes like apathy and irritability. The liver is enlarged due to certain changes. Kwashiorkor is very common among children of 1-3 years of age. The children may have respiratory infections and watery diarrhea (loose motions) and associated deficiencies of vitamins like vitamin A

SYMPTOMS

- i) **Oedema:** Oedema is excessive accumulation of serous fluid in the intercellular spaces of tissues. Oedema is usually observed on the lower limbs, but it may spread to other parts of the body. Oedema is detected by pressing on the skin over the shin (The front part of the leg located below the knee and above the ankle) with your fingers. Because of accumulation of fluid, on applying pressure, there will be a depression at the site of pressure, which will disappear after a minute or so.
- ii) **Growth Failure:** Growth failure is identified by measuring body weight, which is low for a given age in spite of the oedema. Children with kwashiorkor weigh about 60-70% of the weight of a healthy normal of corresponding age. For example, a 2-year-old child weighs about 12 kg. On the other hand, a child with kwashiorkor may weigh only about 60% of 12 kg i.e. 7 kg. In other words, kwashiorkor children are much lighter than the healthy normal counterparts of the same age.

iii) **Mental Changes:** Children suffering from kwashiorkor are irritable and have no interest in the surroundings.

iv) **Skin Changes:** In addition to the above important signs, there may be typical skin changes like scaly pigmentation (skin covered either fully or partially with scales). The skin may peel off easily, leaving sores and cracks.

v) **Hair Changes:** The hair may be sparse and coppery red in colour. Tufts of hair can be pulled off easily even by simple passing of hands through hair, leaving patches of alopecia (loss of hair).

vi) **Moon face:** The face of the child suffering from kwashiorkor may appear puffy with sagging cheeks and swollen eyelids. This sign is normally known as moon face.

vii) **Associated infections:** The kwashiorkor child is often brought to the doctor/hospital with watery diarrhea (frequent loose motions) and severe respiratory infection (cough). The children with kwashiorkor, who had not been immunized, often may be suffering from measles, a childhood disease, characterized by skin rash and fever. However, now a day, due to successful universal immunization program, measles is not as common.

TREATMENT

- Gradually increase calorie intake through small, frequent meals.
- Provide liquid protein supplements.
- Multivitamin supplements and medications to improve appetite.
- Hospitalization may be necessary for severe cases.

ROLE OF GOVERNMENT AND NGOS:

- Promote food security programs.
- Provide nutrition education and counselling.
- Ensure access to healthcare services, including treatment for malnutrition.

MARASMUS

Marasmus is a condition characterized by very low body weight for age, loss of subcutaneous fat (fat under the skin), gross muscle wasting. It is observed more frequently in infants and very young children.

Some common clinical features of marasmus include:

i) Muscle Wasting: The characteristic sign of marasmus is the extensive wasting of muscle with little or no fat under the skin. We use the term wasting to mean emaciation or thinness of the body. The ribs become very prominent. Because of the absence of fat, the skin will develop a number of folds, particularly on the buttocks.

ii) Failure to thrive: There is failure to thrive and the child suffering from marasmus usually is irritable and fretful. In fact, the child is often so weak that the cry of the child cannot even be heard.

iii) Growth failure: Failure to grow is another important feature of the disease. The children often weigh about 50 per cent or less of normal children for their age. For example, a healthy normal one year old child weighs about 10 kg, whereas, a marasmus child would weigh only about 5 to 6 kg.

TREATMENT

- Similar to kwashiorkor, gradually increase calorie and protein intake.
- May require liquid protein supplements.
- Multivitamin supplements and medications to improve appetite.
- Hospitalization may be necessary for severe cases.

ROLE OF GOVERNMENT AND NGOS:

Same as for kwashiorkor, with a focus on food security, nutrition education, and healthcare access.

VITAMIN A DEFICIENCY

Vitamin A deficiency, characterized by symptoms like night blindness and dry eyes.

Symptoms of Vitamin A Deficiency:

- **Night blindness:** Difficulty seeing in low light.
 - **Dry eyes (xerophthalmia):** Eyes feel dry, irritated, or unable to produce enough tears.
 - **Bitot's spots:** Foamy, white spots on the whites of the eyes.
 - **Increased risk of infections:** Weakened immune system.
 - **Delayed growth and development:** In children.
 - **Dry, rough skin:**
 - **Eye inflammation:** Affecting eyelids, surrounding tissues, and even eyes.
- Treatment:
- **Mild deficiency:** Eating vitamin A-rich foods like liver, beef, oily fish, chicken, eggs, fortified milk, carrots, mangoes, sweet potatoes, and leafy green vegetables.
 - **Severe deficiency:** High-dose vitamin A supplements, potentially intravenously if needed.
 - **Diagnosis:** Clinical assessment, medical history, and laboratory tests like electroretinography and serum retinol concentration tests.

ROLE OF GOVERNMENT AND NGOS IN PREVENTION:

Role of Government

National programs for vitamin A supplementation, food fortification, and health education.

- **Role of NGOs:**

Implementing community-based programs, providing education on nutrition, and distributing supplements.

- **Prevention Strategies:**

- **Dietary Diversification:** Encouraging consumption of vitamin, A-rich foods.
- **Food Fortification:** Adding vitamin, A to staple foods like milk and cereals.
- **Breastfeeding Promotion:** Supporting exclusive breastfeeding for the first six months.
- **Periodic Vitamin A Supplementation:** Distributing high-dose supplements to children and pregnant women.

Nutrient Deficiency Control program, namely, National Prophylaxis Program for Prevention of Blindness due to Vitamin A deficiency, National Nutritional Anemia Prophylaxis Program National Iodine Deficiency Disorder Control Program.

IRON DEFICIENCY DISORDERS

Iron deficiency is defined as a condition in which there are no mobilizable iron stores and in which signs of a compromised supply of iron to tissues, including the erythron, are noted. The more severe stages of iron deficiency are associated with anemia.

Iron deficiency disorders, like anemia, cause fatigue, weakness, and pale skin. Treatment involves iron supplements and addressing underlying causes. Government and NGOs play a vital role in prevention through education, fortification of foods, and supplementation programs, especially targeting vulnerable groups like pregnant women and children.

SYMPTOMS

- **Iron-deficiency anemia:**

Fatigue, weakness, shortness of breath, pale skin, and, in severe cases, headaches, dizziness, and impaired cognitive function.

- **Other Iron Deficiency Symptoms:**

Reduced energy levels, decreased work capacity, impaired cognitive and motor development in children, and increased risk of infections.

TREATMENT

- **Iron Supplements:**

Oral or intravenous iron supplements are commonly used to replenish iron stores.

- **Addressing Underlying Causes:**

If iron deficiency is caused by a specific condition (like bleeding disorders or malabsorption), treating that condition is crucial.

- **Dietary Changes:**

Increasing iron-rich foods in the diet can help, especially for preventing deficiency.

ROLE OF GOVERNMENT AND NGOS IN PREVENTION:

- **Fortification:** Adding iron to staple foods like flour or rice to increase iron intake.
- **Supplementation Programs:** Providing iron supplements to vulnerable groups, such as pregnant women and children.
- **Health Education:** Raising awareness about iron deficiency and its prevention through campaigns and community outreach.
- **Anemia Mukt Bharat Initiative:** A government initiative in India to address anemia, focusing on six target groups with various interventions.
- **Mother-Infant Surveillance Systems:** Tracking and monitoring pregnant women and infants for anemia.
- **Community Outreach:** Educating communities about iron deficiency and its prevention.
- **Nutritional Support:** Providing nutritional counseling and resources to individuals and families.
- **Advocacy:** Lobbying for policies that address iron deficiency and promote access to healthy food and healthcare.
- **Collaboration with Government:** Partnering with government agencies to implement prevention programs and improve access to healthcare.

IODINE DEFICIENCY DISORDERS

Iodine Deficiency Disorders (IDDs) are a group of health problems caused by insufficient iodine intake. Symptoms can range from mild to severe, including goiter, mental retardation, and cretinism. Treatment involves iodine supplementation, and prevention is primarily achieved through fortification of salt with iodine. Government and NGOs play crucial roles in implementing and supporting IDD control programs.

SYMPTOMS

- **Goiter:** Swelling in the neck due to an enlarged thyroid gland.
- **Mental Retardation:** Impaired cognitive function, particularly in severe cases of deficiency.
- **Cretinism:** A severe form of IDD affecting physical and mental development, leading to stunted growth, deafness, and other neurological issues.
- **Slow Physical Development:** Reduced growth rate and delayed sexual maturation.
- **Poor School Performance:** Difficulty in learning and academic achievement.
- **Hypothyroidism:** Symptoms like cold intolerance, weight gain, fatigue, and dry skin can be experienced.
- **Other:** Hair loss, dry skin, and puffiness can also be associated with iodine deficiency.

Treatment of Iodine Deficiency Disorders:

- **Iodine Supplementation:**

Supplementation with iodine through iodized salt or oral supplements can help restore iodine levels.

- **Thyroid Hormone Therapy:**

In severe cases, thyroid hormone supplementation may be necessary, especially in infants with cretinism.

ROLE OF GOVERNMENT AND NGOS IN PREVENTION

- **Government Programs:**

National programs like the National Iodine Deficiency Disorders Control Program (NIDDCP) aim to prevent IDD through various strategies, including:

- **Fortification of Salt:** Mandating iodized salt at the production and consumption levels.
- **Health Education:** Promoting awareness about the importance of iodized salt and the dangers of iodine deficiency.
- **Monitoring and Surveys:** Conducting surveys to assess iodine status and monitor the effectiveness of IDD control programs.
- **Training and Capacity Building:** Providing training to health workers and other stakeholders on IDD prevention.
- **NGO Involvement:**
Non-governmental organizations play a vital role in:
- **Community Mobilization:** Working at the grassroots level to raise awareness and promote the consumption of iodized salt.
- **Health Education and Outreach:** Conducting health education campaigns and providing counseling on nutrition and iodine intake.

- **Supporting Government Programs:** Assisting in the implementation of government IDD control programs and providing technical support.
- **Advocacy and Policy Influence:** Advocating for policies that promote iodine fortification and support IDD control efforts.

Example of Government and NGO Collaboration:

- **NIDDCP:**

The National Iodine Deficiency Disorders Control Program in India works in collaboration with various NGOs to ensure the successful implementation of its goals.

- **Community Health Workers:**

NGOs often collaborate with community health workers to reach remote areas and provide health education and support.

- **Public-Private Partnerships:**

Government and NGOs can also collaborate with private sector companies involved in salt production to ensure the availability of iodized salt in various regions.

Iodine deficiency is a preventable public health issue. By combining government programs with the active participation of NGOs, it's possible to significantly reduce the burden of IDD and improve the health and well-being of individuals and communities.

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UNIT IV: LIFE STYLE DEPENDENT DISEASES

Hypertension, Diabetes mellitus, and Obesity their causes prevention. Social health problems- Smoking, Alcoholism, Narcotics. Acquired Immune Deficiency Syndrome (AIDS): causes, treatment and prevention.

HYPERTENSION

Hypertension or high blood pressure is a serious health problem which currently affects nearly 1 billion people worldwide. According to the recent analysis by the WHO, this statistic might rise to around 1.57 billion by the year 2025.

CAUSES

Unhealthy Diet:

Excessive sodium intake, a diet high in saturated and trans fats, and low intake of fruits and vegetables can contribute to hypertension.

Lack of Physical Activity:

Sedentary lifestyles are linked to an increased risk of developing high blood pressure.

Obesity:

Being overweight or obese increases the strain on the heart and blood vessels, raising the risk of hypertension.

Tobacco and Alcohol Use:

Smoking and excessive alcohol consumption can damage blood vessels and raise blood pressure.

Age and Family History:

Hypertension is more common in older individuals and those with a family history of the condition.

Stress:

Chronic stress can lead to temporary spikes in blood pressure and contribute to the long-term development of hypertension.

Underlying Health Conditions:

Conditions like kidney disease, diabetes, and sleep apnea can increase the risk of hypertension.

Pregnancy:

High blood pressure can occur during pregnancy, often leading to complications like preeclampsia.

PREVENTION

- **Healthy Diet:** Consume a balanced diet rich in fruits, vegetables, whole grains, and lean proteins while limiting sodium, saturated fat, and trans fats.

- **Regular Physical Activity:** Aim for at least 150 minutes of moderate-intensity exercise per week.
- **Weight Management:** Maintain a healthy weight to reduce the strain on the heart and blood vessels.
- **Stress Management:** Practice stress-reducing techniques like meditation, yoga, or deep breathing.
- **Limit Alcohol Consumption:** If you drink alcohol, do so in moderation.
- **Quit Smoking:** Smoking damages blood vessels and increases the risk of hypertension.
- **Manage Underlying Conditions:** If you have diabetes, kidney disease, or other conditions that increase your risk of hypertension, work with your doctor to manage them effectively.
- **Regular Checkups:** Get your blood pressure checked regularly to identify and manage any potential issues early.

DIABETES MELLITUS

Diabetes mellitus is a chronic metabolic disorder characterized by high blood sugar levels. It's caused by either the body's inability to produce insulin or the body's resistance to insulin, leading to a buildup of glucose in the bloodstream.

CAUSES

1. **Insulin resistance:** Type 2 diabetes mainly results from insulin resistance. Insulin resistance happens when cells in your muscles, fat and liver don't respond as they should to insulin. Several factors and conditions contribute to varying degrees of insulin resistance, including obesity, lack of physical activity, diet, hormonal imbalances, genetics and certain medications.
2. **Autoimmune disease:** Type 1 diabetes and LADA happen when your immune system attacks the insulin-producing cells in pancreas.
3. **Hormonal imbalances:** During pregnancy, the placenta releases hormones that cause insulin resistance. You may develop gestational diabetes if your pancreas can't produce enough insulin to overcome the insulin resistance. Other hormone-related conditions like acromegaly and Cushing syndrome can also cause Type 2 diabetes.
4. **Pancreatic damage:** Physical damage to your pancreas — from a condition, surgery or injury — can impact its ability to make insulin, resulting in Type 3c diabetes.
5. **Genetic mutations:** Certain genetic mutations can cause MODY and neonatal diabetes.

SYMPTOMS

Symptoms of high blood sugar include:

- a. Increased thirst
- b. Increased urination
- c. Increased hunger

When blood glucose is higher than 160 to 180 mg/dL (8.9 to 10.0 mmol/L), glucose appears in the urine. When the level of glucose in the urine rises further, the kidneys excrete more water to dilute a large amount of glucose. Because the kidneys make too much urine, people with diabetes urinate a lot and often (polyuria). Excessive urination creates abnormal thirst (polydipsia). Once the excess calories are lost in the urine, the person may lose weight. To compensate, the person often feels excessively hungry.

Other symptoms of diabetes include

- a. Blurry vision
- b. Drowsiness
- c. Nausea
- d. Decreased resistance during physical activity

PREVENTION

- **Maintain a Healthy Weight:**

Losing even a small amount of weight can significantly reduce the risk of developing type 2 diabetes.

- **Engage in Regular Physical Activity:**

Aim for at least 150 minutes of moderate-intensity exercise per week.

- **Follow a Balanced Diet:**

Eat a diet rich in fruits, vegetables, and whole grains, and limit sugary drinks and processed foods.

- **Stop Smoking:**

Smoking can increase the risk of developing type 2 diabetes and can also worsen the condition.

- **Manage Other Risk Factors:**

Control high blood pressure and high cholesterol, as these can increase the risk of diabetes.

- **Regular Checkups:**

Get regular checkups with your doctor, especially if you have a family history of diabetes or other risk factors.

OBESITY

Obesity, defined as excessive body fat accumulation, is a major health concern with complex causes. It arises from an imbalance between energy intake and expenditure, often influenced by genetics, environment, and lifestyle choices.

CAUSES

1. Energy Imbalance:

Consuming more calories than the body burns through daily activities and exercise leads to weight gain and fat storage.

2. Dietary Factors:

A diet high in calories, fat, processed foods, and sugary drinks contributes to obesity.

3. Inactivity:

A sedentary lifestyle, where individuals spend little time engaging in physical activity, hinders calorie burning and promotes weight gain.

4. Genetics:

Family history and genetic predispositions can increase the likelihood of developing obesity.

5. Environmental Factors:

Access to affordable unhealthy foods, lack of safe spaces for physical activity, and fast-paced lifestyles can contribute to obesity.

6. Medical Conditions and Medications:

Some medical conditions (like Cushing's syndrome, hypothyroidism) and medications (like some antidepressants and diabetes drugs) can lead to weight gain.

7. Psychological Factors:

Stress, depression, and other mental health issues can influence eating habits and contribute to obesity.

8. Sleep Deprivation:

Lack of sleep can disrupt hormone regulation, increase appetite and decrease energy expenditure.

Prevention and Management of Obesity:

9. Healthy Eating:

Focus on a balanced diet with plenty of fruits, vegetables, lean protein, and whole grains. Limit processed foods, sugary drinks, and unhealthy fats.

10. Regular Physical Activity:

Engage in regular exercise, aiming for at least 150 minutes of moderate-intensity aerobic activity per week.

11. Stress Management:

Implement healthy stress management techniques like yoga, meditation, or spending time in nature.

12. Adequate Sleep:

Prioritize sufficient sleep to regulate hormones and improve metabolic health.

13. Behavioural Changes:

Identify and modify unhealthy eating and activity patterns. This may involve setting realistic goals, keeping a food journal, and seeking support from friends or family.

14. Professional Guidance:

Consult with a doctor, registered dietitian, or certified personal trainer for personalized advice and support.

15. Addressing Medical Conditions:

If obesity is linked to a medical condition, work with a healthcare professional to manage the underlying cause.

Social health problems- Smoking, Alcoholism, Narcotics.

SMOKING

Smoking refers to the inhalation of the smoke of burning tobacco, commonly through cigarettes, bidis, cigars, or pipes.

Health Effects

- Lung cancer, chronic bronchitis, and emphysema
- Increased risk of heart disease and stroke
- Weakening of the immune system
- Reduced oxygen supplies due to carbon monoxide
- Harmful effects on passive smokers, especially children and pregnant women

Social Effects

- Financial burden on families
- Reduced productivity and increased absenteeism
- Risk of second-hand smoke exposure in public spaces
- Social stigma in many cultures and workplaces

Control Measures

- Public awareness campaigns
- Warning labels on cigarette packs
- Smoking bans in public places
- Counselling and rehabilitation programs
- Increased taxation on tobacco products

ROLE OF GOVERNMENT

Regulation:

1. COTPA (Cigarettes and Other Tobacco Products Act, 2003): This Act prohibits direct and indirect tobacco product advertising, bans the sponsorship of events by tobacco companies, and restricts the sale of tobacco products to minors.

2. Smoke-free environments: Bans on smoking in public places and enclosed workplaces protect non-smokers from secondhand smoke.

Public Health:

1. NTCP (National Tobacco Control Program): Aims to reduce tobacco use through awareness campaigns, cessation programs, and enforcement of tobacco control laws.

2. Global Adult Tobacco Survey (GATS): Provides data on tobacco use prevalence, helping to track progress and inform policy.

3. Publicity and awareness campaigns: Utilize various media platforms to educate the public about the dangers of smoking and promote cessation.

ALCOHOLISM

Alcoholism is the chronic and compulsive consumption of alcoholic beverages, leading to addiction and health deterioration.

Health Effects

- Liver damage (cirrhosis, fatty liver)
- Cardiovascular issues
- Impaired brain function and memory loss
- Mental health issues like depression and anxiety
- Risk of accidents due to impaired judgment

Social Effects

- Family violence and marital discord
- Workplace issues and unemployment
- Road accidents and criminal behaviour
- Neglect of responsibilities and children
- Increased healthcare burden

Control Measures

- Regulation of alcohol sales and advertisements
- Alcohol education in schools and colleges
- Community-based intervention programs
- Access to de-addiction centres and therapy
- Legal age limits for consumption

ROLE OF GOVERNMENT

The Ministry of Social Justice and Empowerment has introduced Scheme of Assistance for Prevention of Alcoholism and Substance (Drugs) Abuse and for Social Defence Services. The scheme seeks to benefit victims of alcohol and substance abuse. Alpha Healing Centre in India is elevated with a purpose to provide world-class inpatient treatment centre for alcohol addiction, with the help of highly qualified counsellors and unique treatment programs.

NARCOTICS (DRUG ABUSE)

Narcotics are drugs that dull the senses and relieve pain, but many are abused for their euphoric effects. Common narcotics include heroin, opium, morphine, and synthetic drugs.

Health Effects

- Physical dependence and withdrawal symptoms
- Nervous system damage
- Risk of infectious diseases (e.g., HIV/AIDS, hepatitis from needle sharing)
- Mental disorders (e.g., paranoia, psychosis)
- Risk of overdose and death

Social Effects

- Increase in crime (theft, violence, trafficking)
- Family breakdown and social isolation
- Dropout from education and work
- Burden on healthcare and legal systems
- Stigma and discrimination

Control Measures

- Strict drug control laws and enforcement (e.g., NDPS Act in India)
- School and college drug awareness programs
- Counselling, detox, and rehabilitation services
- Community vigilance and youth support groups
- International cooperation to combat trafficking

ROLE OF GOVERNMENT

The government's role in narcotics typically revolves around preventing drug trafficking and abuse through various strategies, including law enforcement, policy formulation, and public awareness campaigns.

1. LAW ENFORCEMENT AND DRUG CONTROL:

- **Enforcing narcotics laws:**

Governments establish and enforce laws that prohibit the production, trafficking, and use of illegal narcotics.

- **Combating drug trafficking:**

They establish specialized agencies, like the Narcotics Control Bureau in India, to target drug trafficking networks and seize illicit substances.

- **Border control and monitoring:**

Governments implement measures to monitor borders and prevent the smuggling of drugs into the country.

- **International cooperation:**

Governments work with other nations to share information, collaborate on joint operations, and address cross-border drug trafficking.

2. POLICY AND LEGISLATION

- **Formulating and enacting narcotics laws:**

Governments develop and implement laws related to the control and regulation of drugs.

- **Setting drug policy:**

Governments establish policies on drug use, treatment, and prevention, including approaches to decriminalization, harm reduction, and rehabilitation.

- **Monitoring and evaluating drug policies:**

Governments assess the effectiveness of their drug policies and make adjustments as needed.

3. PUBLIC AWARENESS AND PREVENTION

- **Public education campaigns:**

Governments often run campaigns to educate the public about the dangers of drug abuse, including information on addiction, treatment options, and prevention strategies.

- **Support for treatment and rehabilitation programs:**

Governments may provide funding or support for treatment and rehabilitation programs for individuals struggling with drug addiction.

- **Prevention programs in schools and communities:**

Governments may invest in prevention programs that target young people and communities at risk of drug abuse.

4. COORDINATION AND COLLABORATION

- **Collaboration between government agencies:**

Governments often involve multiple agencies in narcotics control, including police, customs, intelligence, and health departments.

- **Partnerships with NGOs and community organizations:**

Governments may work with NGOs and community organizations to address drug abuse, provide support services, and raise awareness.

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UNIT V: DISEASE CAUSED BY MICROORGANISM

Food and water born infection: Bacterial diseases, Cholera, Typhoid fever. Viral diseases Hepatitis, Polio myelitis. Protozoan diseases: amoebiasis, giardiasis - Parasitic diseases: taeniasis and ascariasis their transmission, causative agent, sources of infection, symptoms and prevention. Causes of food spoilage and its prevention.

FOOD AND WATER BORNE DISEASES

These are illnesses caused by consuming contaminated food or drinking water. These diseases are commonly spread through bacteria, viruses, parasites, or toxins present in unclean food or water. Poor sanitation, improper food handling, and lack of access to safe drinking water are major contributors to their spread.

Common examples include cholera, typhoid, hepatitis A, amoebiasis, and food poisoning. These diseases can lead to serious health issues like diarrhea, dehydration, and malnutrition, especially in children. Preventive measures such as maintaining personal hygiene, ensuring proper cooking and storage of food, and using clean drinking water are essential to control their occurrence.

BACTERIAL DISEASES

Bacterial diseases, also known as bacterial infections, are illnesses caused by bacteria entering the body and causing harm. These infections can range in severity, from mild conditions like food poisoning to life-threatening diseases like sepsis.

CHOLERA

CAUSATIVE AGENT: *Vibrio cholerae* (a gram-negative bacterium)

TRANSMISSION

- Faecal-oral route
- Drinking contaminated water or eating contaminated food

SOURCES OF INFECTION

- Contaminated drinking water
- Raw or undercooked seafood
- Poor sanitation

SYMPTOMS

- Sudden onset of watery diarrhoea ("rice water stools")
- Vomiting
- Dehydration
- Muscle cramps
- If untreated, may lead to shock or death

PREVENTION

- Drink safe, boiled or treated water
- Maintain good sanitation and hygiene
- Proper disposal of human waste
- Wash hands with soap

TYPHOID FEVER

CAUSATIVE AGENT: *Salmonella typhi* (a gram-negative bacterium)

TRANSMISSION

- Faecal-oral route
- Consumption of food or water contaminated with faeces of infected individuals

SOURCES OF INFECTION

- Contaminated water or milk
- Unhygienic food handling
- Carriers (people who recovered but still shed the bacteria)

SYMPTOMS

- Persistent high fever (up to 104°F / 40°C)
- Weakness and fatigue
- Abdominal pain
- Headache
- Constipation or diarrhoea
- Rose-coloured spots on the abdomen (in some cases)

PREVENTION

- Ensure safe drinking water
- Maintain food hygiene
- Wash hands frequently
- Avoid eating uncooked street food
- Typhoid vaccination (for travellers and during outbreaks)

VIRAL DISEASES

Viral diseases, also known as viral infections, are illnesses caused by viruses. These infections can affect various parts of the body, ranging from common colds to serious conditions like AIDS.

HEPATITIS A

CAUSATIVE AGENT: Hepatitis A virus (HAV) – an RNA virus

TRANSMISSION

- **Faecal-oral route**
- Ingestion of food or water contaminated with feces

SOURCES OF INFECTION

- Contaminated drinking water
- Raw or undercooked shellfish
- Poor personal hygiene and sanitation

SYMPTOMS

- Fever
- Fatigue
- Loss of appetite
- Nausea and vomiting
- Jaundice (yellowing of skin and eyes)
- Dark urine and pale stools

PREVENTION

- Drink clean and safe water
- Maintain good hygiene
- Proper sewage disposal
- Hepatitis A vaccine
- Wash hands before eating and after using the toilet

POLIOMYELITIS (POLIO)

CAUSATIVE AGENT: Poliovirus (an enterovirus)

TRANSMISSION

- Faecal-oral route (most common)
- Droplet infection (rarely) through respiratory secretions

SOURCES OF INFECTION

- Contaminated food and water
- Poor sanitation
- Close contact with infected persons

SYMPTOMS

- Most cases: Asymptomatic or mild fever, sore throat
- Some cases:
 - Headache
 - Muscle stiffness
 - Paralysis of limbs (usually lower limbs)
 - Difficulty in breathing (in severe cases)

PREVENTION:

- Maintain proper sanitation
- Ensure safe drinking water
- Polio vaccination (Oral Polio Vaccine – OPV or Inactivated Polio Vaccine – IPV)
- Avoid contact with infected persons

PROTOZOAN DISEASES

Protozoan diseases are illnesses caused by parasitic protozoa, which are single-celled organisms. These diseases can affect humans and other animals, and can range from mild to severe, depending on the specific protozoan and the individual's health. Some common examples of protozoan diseases include malaria, leishmaniasis, and giardiasis.

AMOEBIASIS (AMOEBIIC DYSENTERY)

CAUSATIVE AGENT: *Entamoeba histolytica*

TRANSMISSION:

- Faecal-oral route
- Ingestion of cysts through contaminated food or water

SOURCES OF INFECTION

- Contaminated drinking water
- Unwashed fruits and vegetables
- Flies and cockroaches can carry cysts

SYMPTOMS

- Bloody diarrhoea or loose stools
- Abdominal cramps and pain
- Nausea
- Weakness and fatigue
- In severe cases, may cause liver abscess

PREVENTION:

- Drink boiled or treated water
- Wash hands regularly
- Wash vegetables and fruits thoroughly
- Maintain proper sanitation
- Avoid open defecation

2. GIARDIASIS

- **CAUSATIVE AGENT:** *Giardia lamblia* (also called *Giardia intestinalis*)
- **TRANSMISSION:**
 - Faecal-oral route
 - Ingestion of cysts from contaminated water or food
- **SOURCES OF INFECTION:**
 - Contaminated drinking water (especially untreated surface water)
 - Person-to-person contact (especially in childcare settings)
 - Contaminated fruits and vegetables
- **SYMPTOMS:**
 - Watery and foul-smelling diarrhoea
 - Bloating and gas
 - Nausea and weight loss
 - Abdominal cramps
 - Fatigue
- **PREVENTION:**
 - Drink filtered or boiled water
 - Practice good hand hygiene
 - Avoid drinking untreated water from lakes, rivers, or wells
 - Wash raw produce before consumption

CAUSES OF FOOD SPOILAGE AND PREVENTION

Food spoilage refers to the process by which food becomes unfit for consumption due to physical, chemical, or biological changes. Spoiled food may develop unpleasant taste, odor, texture, or appearance, and in many cases, it can be harmful to health. The main causes of spoilage include microbial growth, enzymatic activity, oxidation, and poor storage conditions. Spoiled food not only leads to wastage but can also pose serious health risks. Therefore, understanding the causes of food spoilage and adopting proper preservation and hygiene practices are essential to ensure food safety and reduce food loss.

1. Microbial Activity

- Caused by bacteria, yeasts, and Molds
- Leads to souring, bad Odor, discoloration, gas formation
- Example: Bread Mold, sour milk

2. Enzymatic Action

- Naturally present enzymes in food can break down components
- Causes softening, browning, and off-Flavors
- Example: Browning of cut fruits (due to polyphenol oxidase)

3. Chemical Reactions

- Oxidation of fats and oils leads to rancidity
- Vitamin loss due to exposure to light and oxygen

4. Physical Changes

- Improper **storage** leads to drying, softening, or cracking
- Freezer burns in frozen foods

5. Insect and Rodent Infestation

- Stored grains can be attacked by insects, rodents, and pests
- Leads to contamination and damage

Prevention of Food Spoilage

1. Proper Storage

- Use refrigeration and freezing to slow microbial growth
- Store dry foods in airtight containers

2. Food Preservation Methods

- Canning, pickling, drying, salting, sugaring
- Use of preservatives like vinegar, salt, sugar

3. Hygiene Practices

- Maintain clean kitchen and utensils
- Wash hands and cover food properly

4. Packaging

- Use vacuum sealing, airtight containers, or modified atmosphere packaging (MAP)

5. Control Temperature & Moisture

- Avoid leaving perishable food at room temperature
- Store in cool, dry places

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